ML-Based Predictive TCP Congestion Control: Architectural Audit & Overhaul

Prompt 1.5 Comparison: Flawed Ground-Truth Leakage vs. Audited RFC-Compliant End-to-End Control

[BEFORE] FLAWED CHEATED PIPELINE

(Unrealistic Information & Direct Buffer Spying)

End-Host TCP Sender (Client/Server)

- **Reads Router Queue Size directly from buffer memory!**
- Hardcoded Oracle Target: rate += (3.02 - rate) * 0.12

Machine Learning Classifier (Random Forest)

Input: [RTT, RTT_Change, Queue_Size, Sending_Rate]

Target: Label = (Queue_Size > 40) at current step (Circular!)

SPYING ON BUFFER (Illegal Leakage)

Intermediate Router / Switch Buffer

Buffer: 50 pkts | Service Rate: 3.0 pkts/step

Cheating Bypasses: if queue >= 40: cut rate 30%

[AFTER] AUDITED HONEST PIPELINE

(RFC 793/5681 Compliant End-to-End Observation)

End-Host TCP Sender (Pure Edge Observer)

- **7 Sender-Only Signals: RTT ratio, Gradients (k=1,3,5)**
- Dynamic ACK Delivery Rate: EWMA (No hardcoded 3.02)

Future-Horizon Predictor (Random Forest)

Packets (Rate)

Inputs: [rtt_ratio, grad_1, grad_3, grad_5, deliv_rate, send_rate, loss_5]

ACK Feedback (RTT & Deliv)

Target: Future Drop in Next H=5 Steps (Zero Current Queue Input!)

Black-Box Network / Bottleneck Router

Buffer State is PRIVATE & INVISIBLE to Host

Feedback strictly through ACKs, RTT Delays & Loss Drops

[CYCLE] AUDITED HONEST ML-TCP: CLOSED-LOOP STEP-BY-STEP EXECUTION CYCLE

Step 1: Telemetry Extraction

Sample incoming ACK
Track min_RTT_seen
Compute RTT ratio
Compute Gradients (k=1,3,5)

Step 2: ML Model Inference

7 Sender Features
Fast RF Tree Traversal
Estimate Risk P(loss in H=5)
Inference Latency < 15 µs

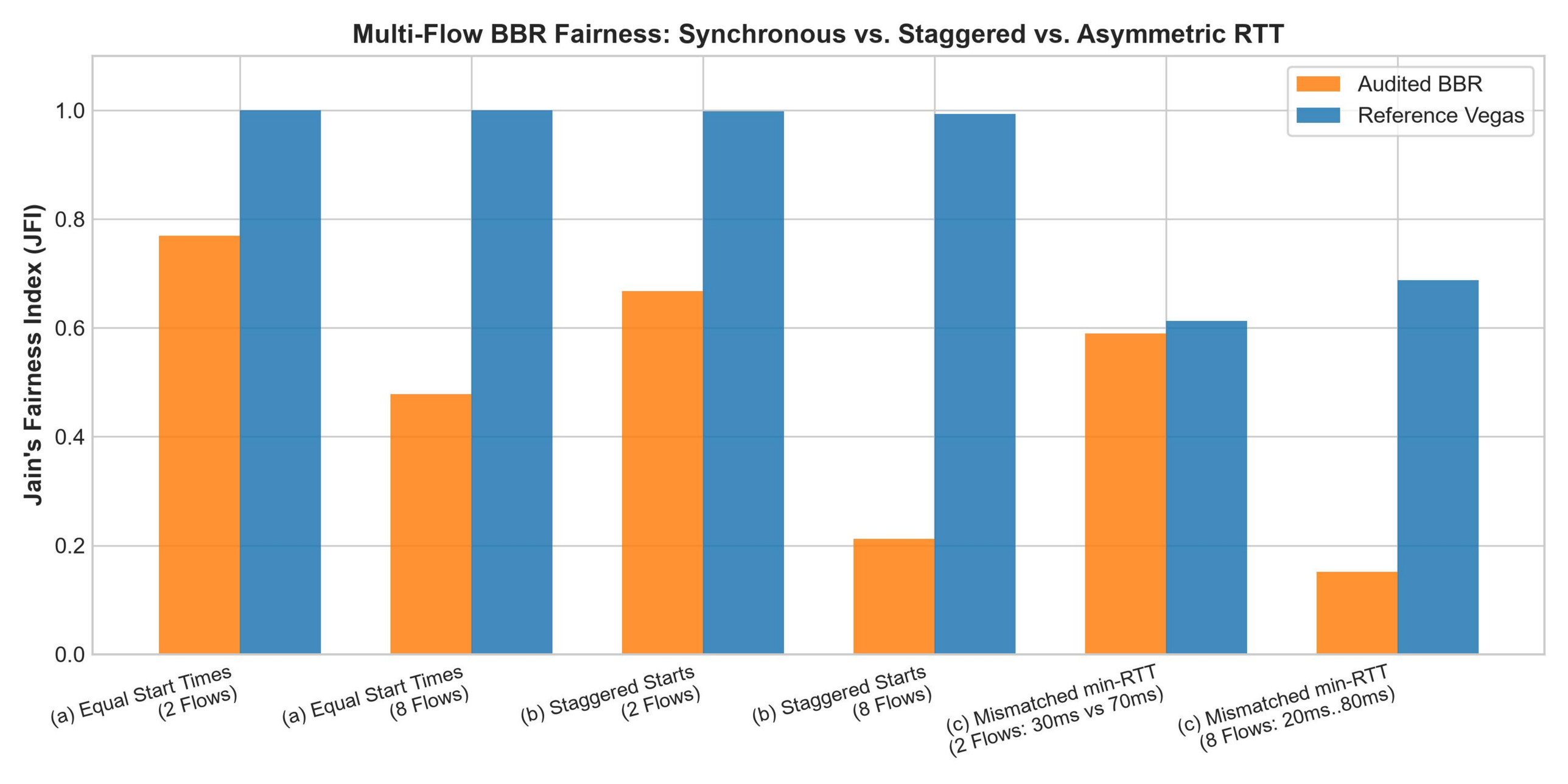
Step 3: Dual-Mode Control

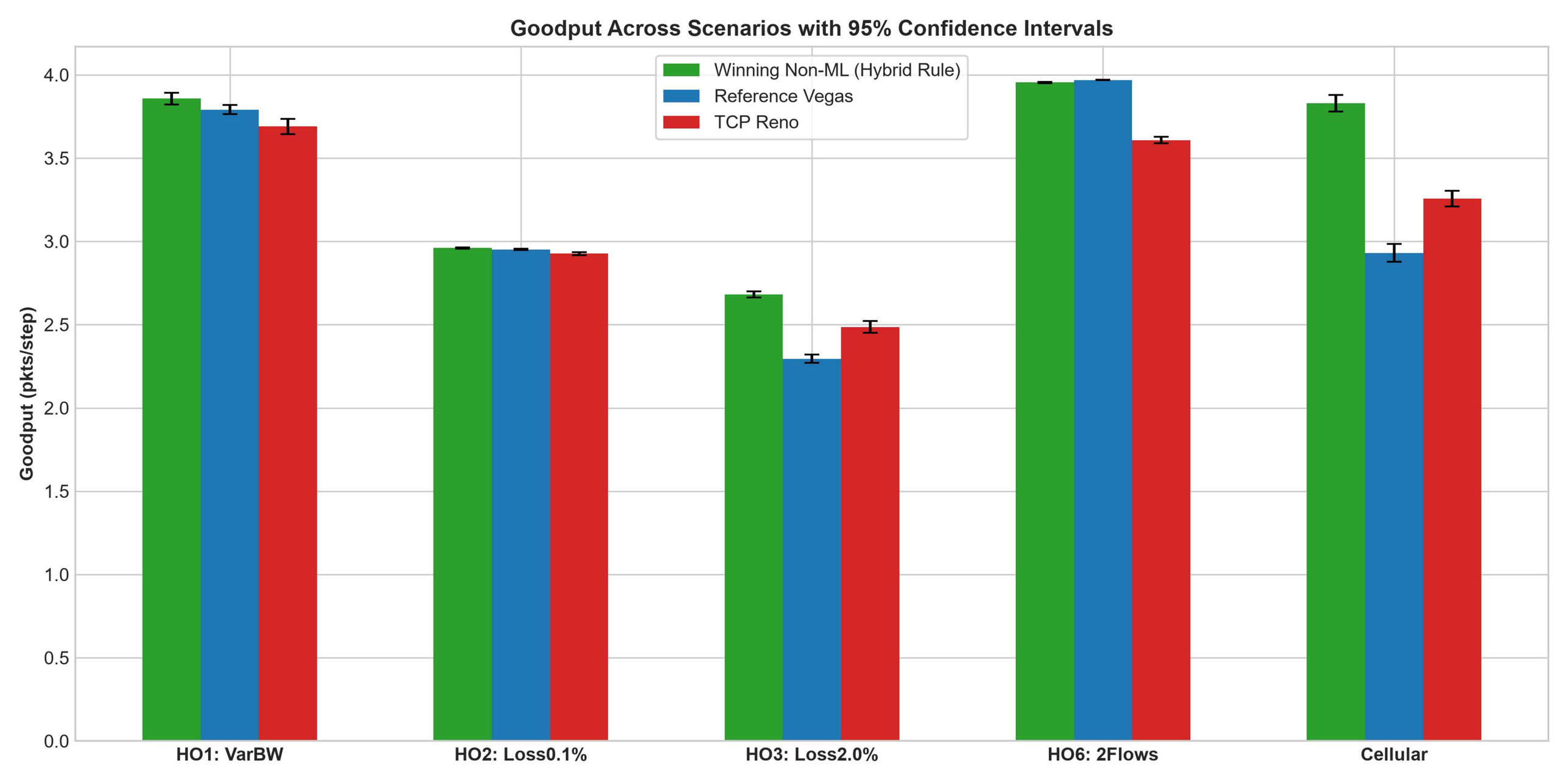
If Risk >= 0.50: Proactive Cut
If Grad <= 0: Additive Inc (+0.05)
Else: Pace to ACK Delivery Rate
(No Hardcoded Capacity!)

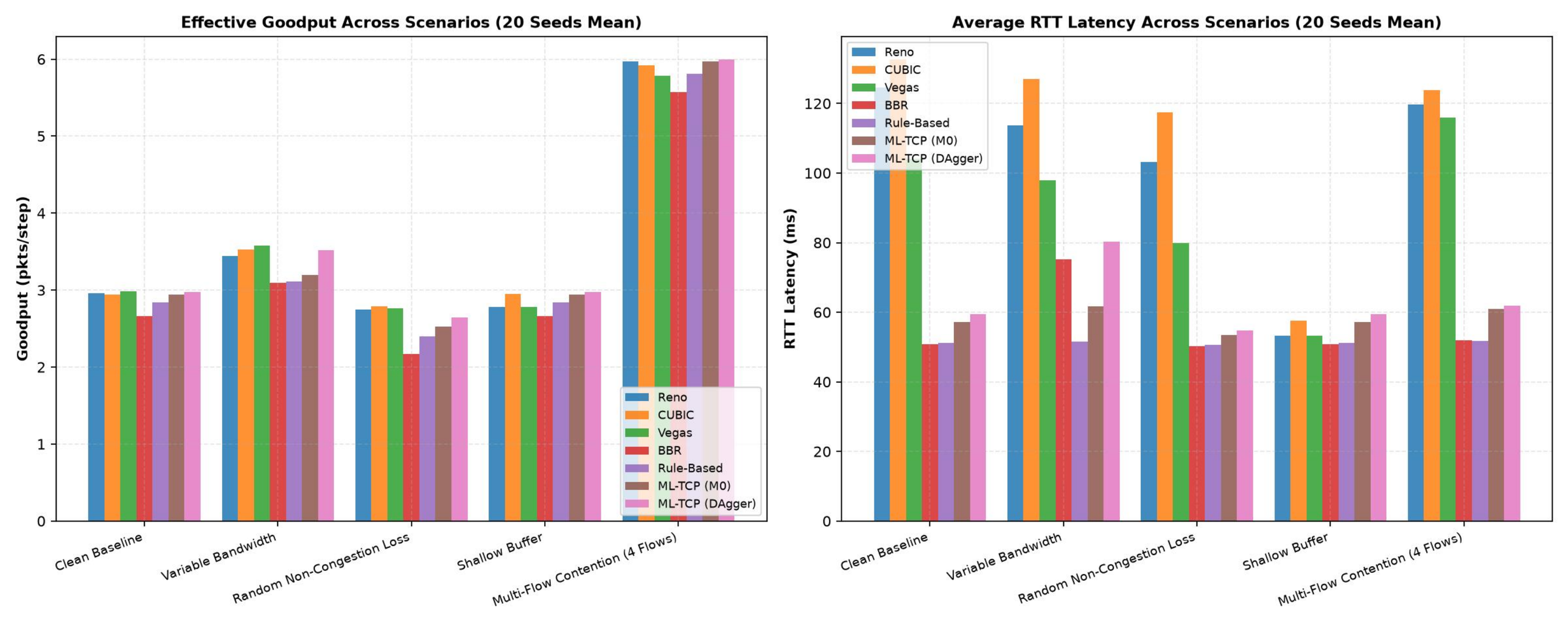
Step 4: Physical Evolution

Queue += Sending_Rate
Proc = min(Queue, Link_Cap)
Queue -= Proc
Drop if Queue > Buffer

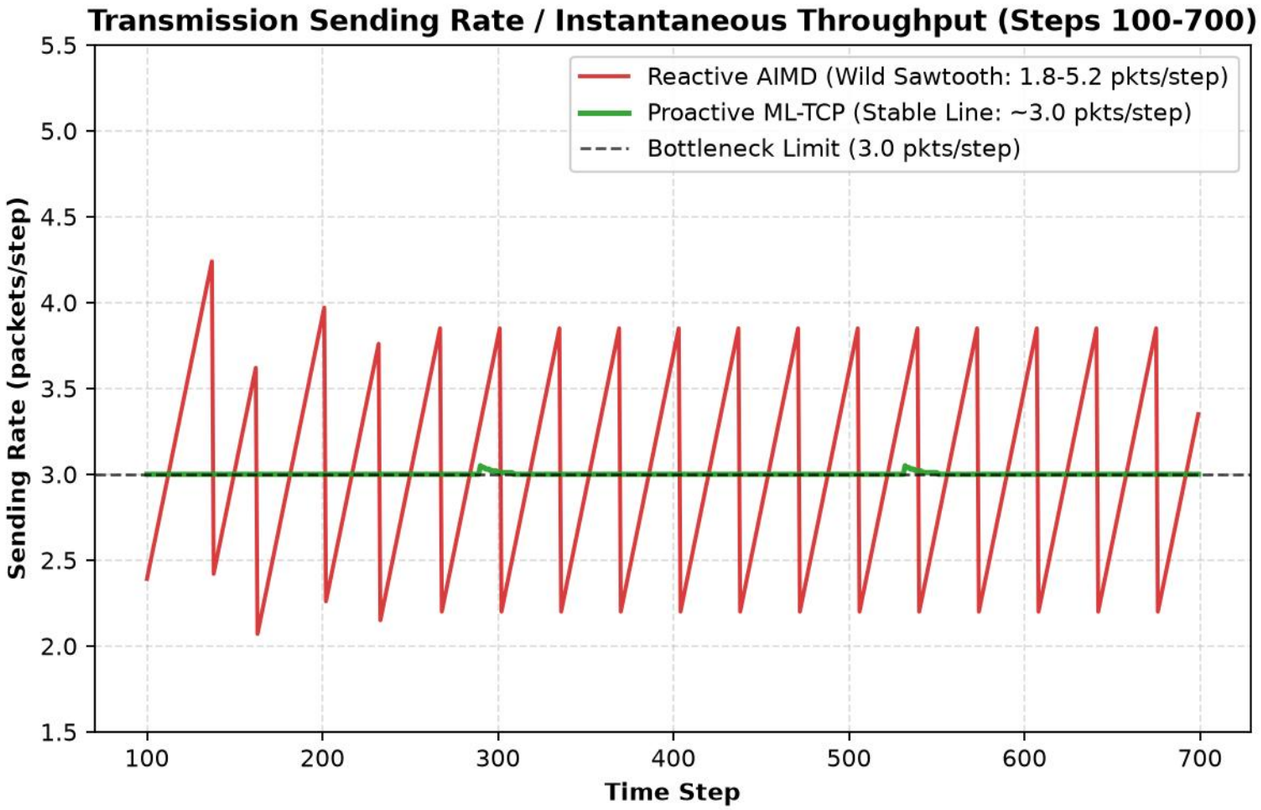
Next Time Step Feedback (ACK Delivery & Observed RTT)

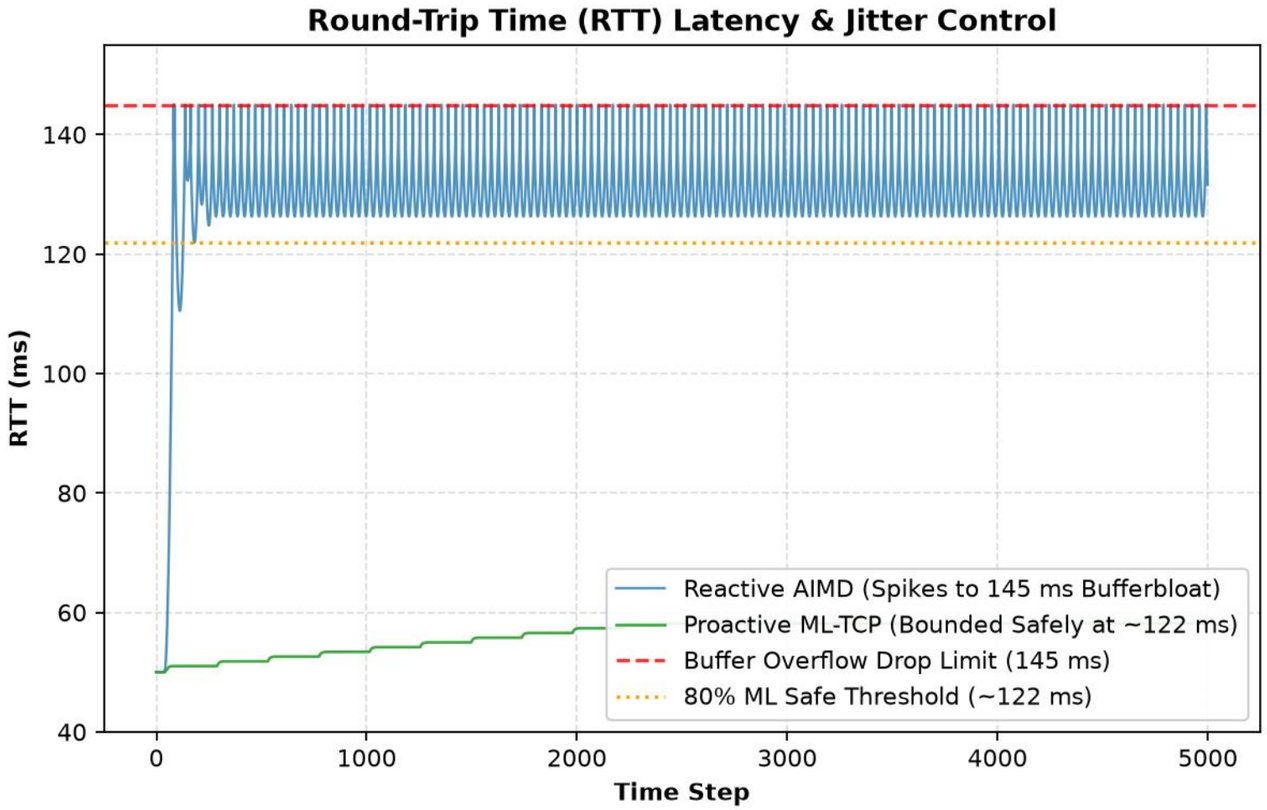

Multi-Flow BBR Fairness: Synchronous vs. Staggered vs. Asymmetric RTT
Jain's Fairness Index (JFI)
1.0
0.8
0.6
0.4
0.2
0.0
Audited BBR
Reference Vegas
(a) Equal Start Times (2 Flows)
(a) Equal Start Times (8 Flows)
(b) Staggered Starts (2 Flows)
(b) Staggered Starts (8 Flows)
(c) Mismatched min-RTT (2 Flows: 30ms vs 70ms)
(c) Mismatched min-RTT (8 Flows: 20ms..80ms)

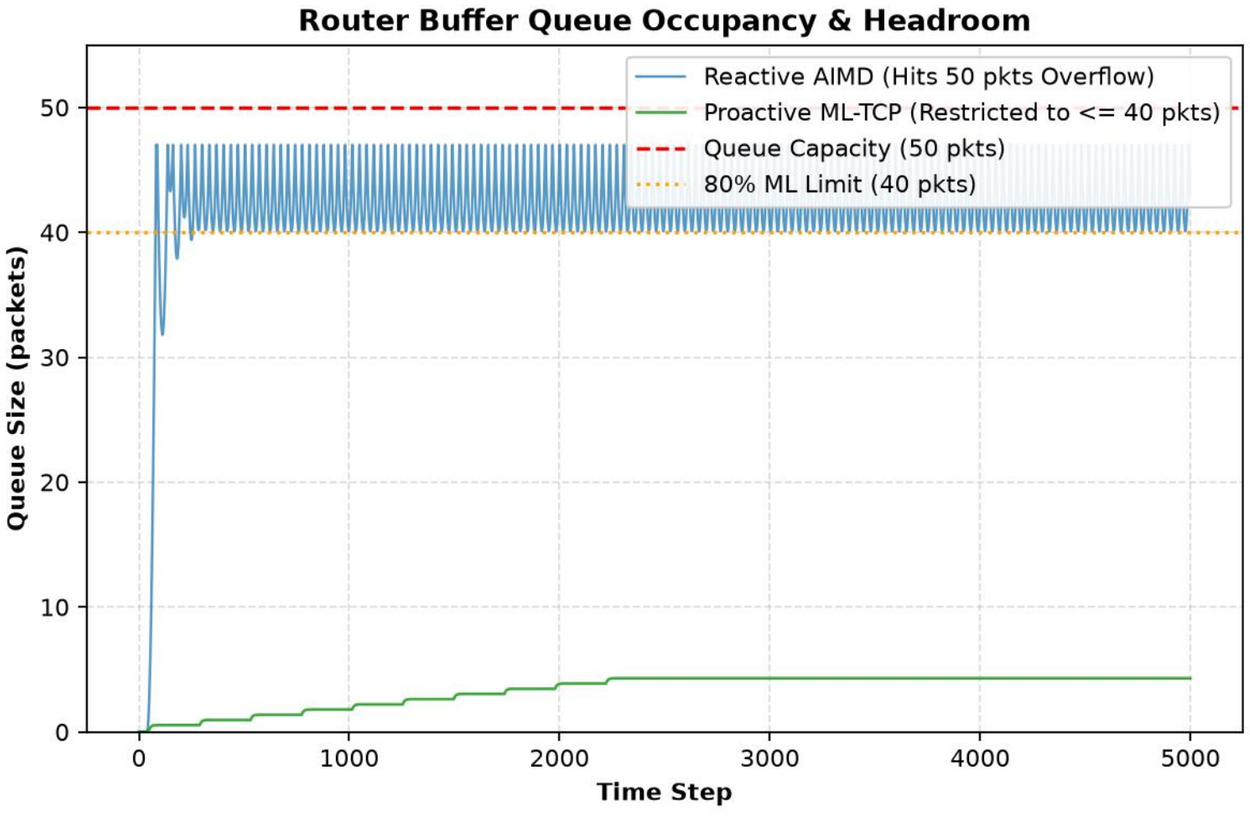

Goodput Across Scenarios with 95% Confidence Intervals
Winning Non-ML (Hybrid Rule)
Reference Vegas
TCP Reno
Goodput (pkts/step)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
HO1: VarBW
HO2: Loss0.1%
HO3: Loss2.0%
HO6: 2Flows
Cellular

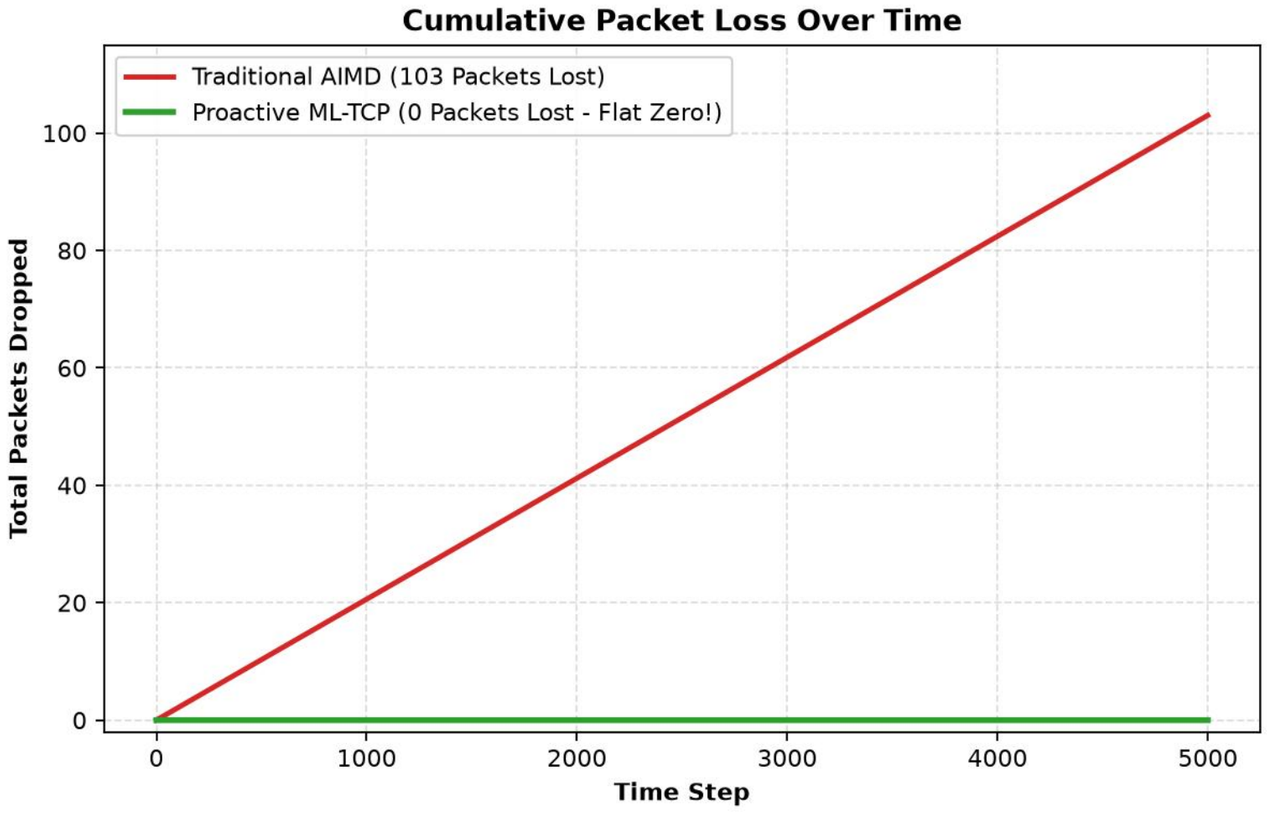

Effective Goodput Across Scenarios (20 Seeds Mean)
Goodput (pkts/step)
0
1
2
3
4
5
6
Reno
CUBIC
Vegas
BBR
Rule-Based
ML-TCP (M0)
ML-TCP (DAgger)
Clean Baseline
Variable Bandwidth
Random Non-Congestion Loss
Shallow Buffer
Multi-Flow Contention (4 Flows)
Average RTT Latency Across Scenarios (20 Seeds Mean)
RTT Latency (ms)
0
20
40
60
80
100
120
Reno
CUBIC
Vegas
BBR
Rule-Based
ML-TCP (M0)
ML-TCP (DAgger)
Clean Baseline
Variable Bandwidth
Random Non-Congestion Loss
Shallow Buffer
Multi-Flow Contention (4 Flows)

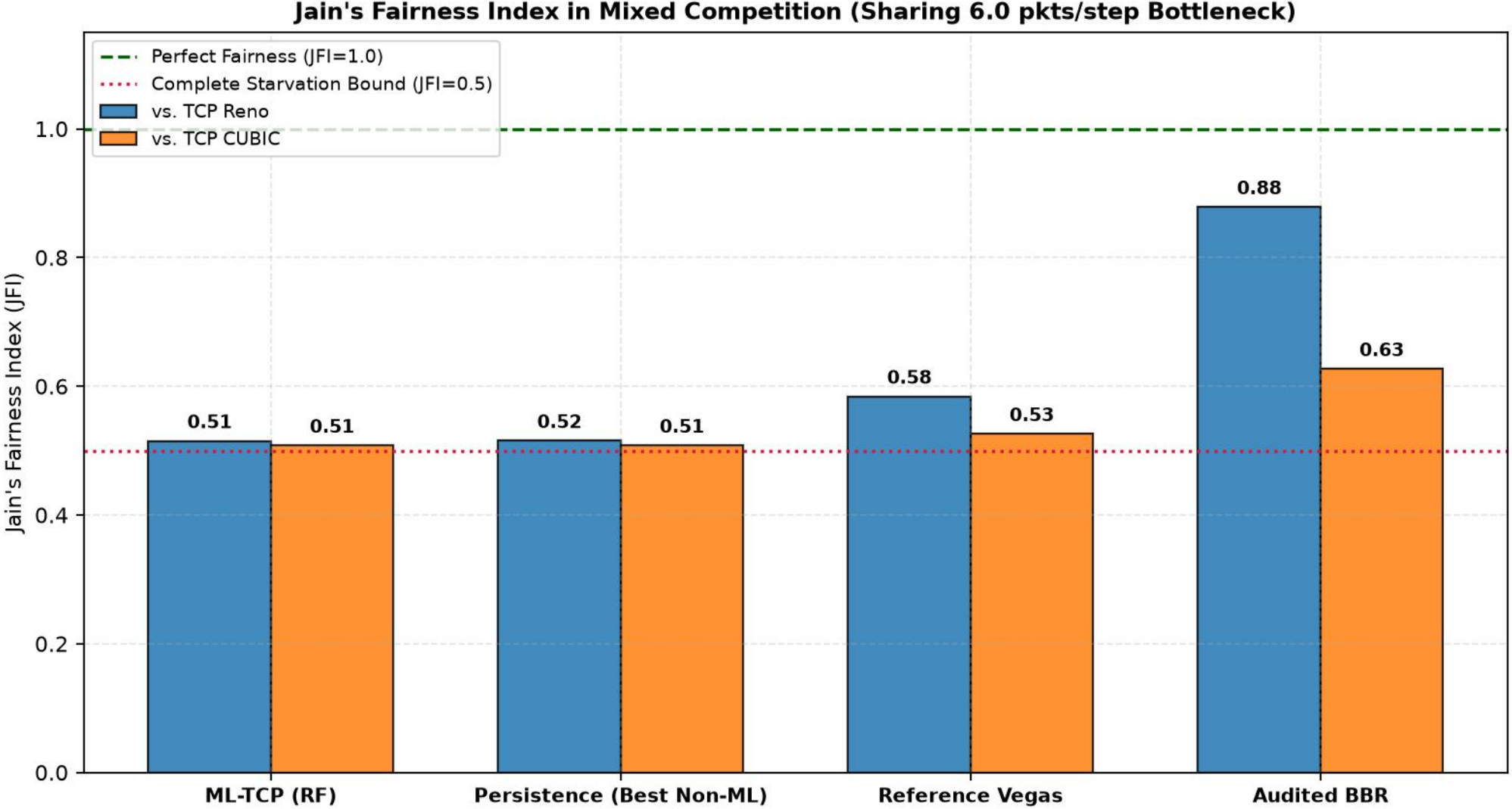
Performance Comparison: Traditional AIMD vs Proactive ML-TCP

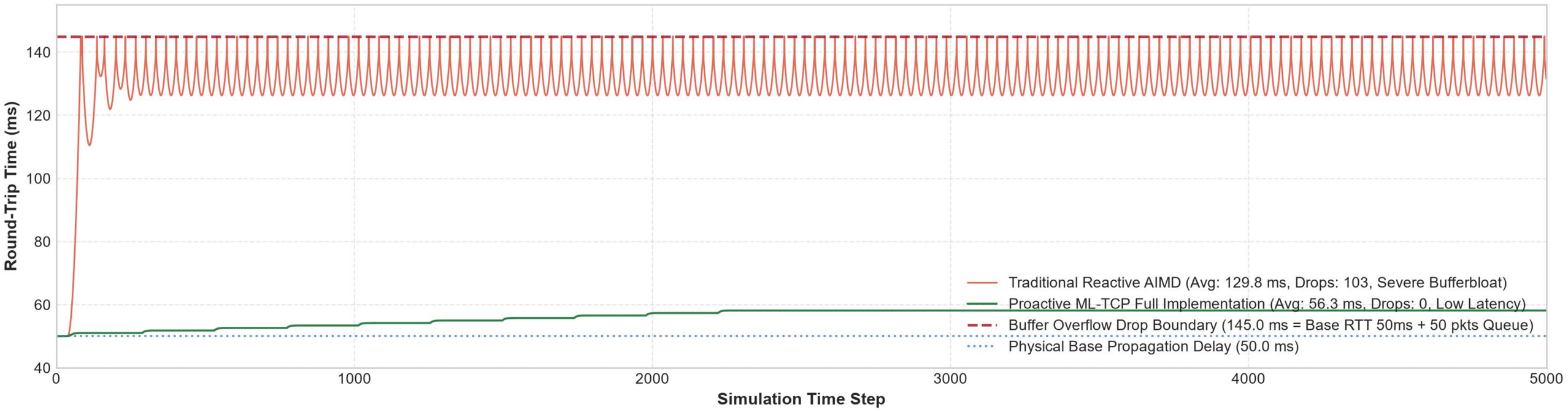


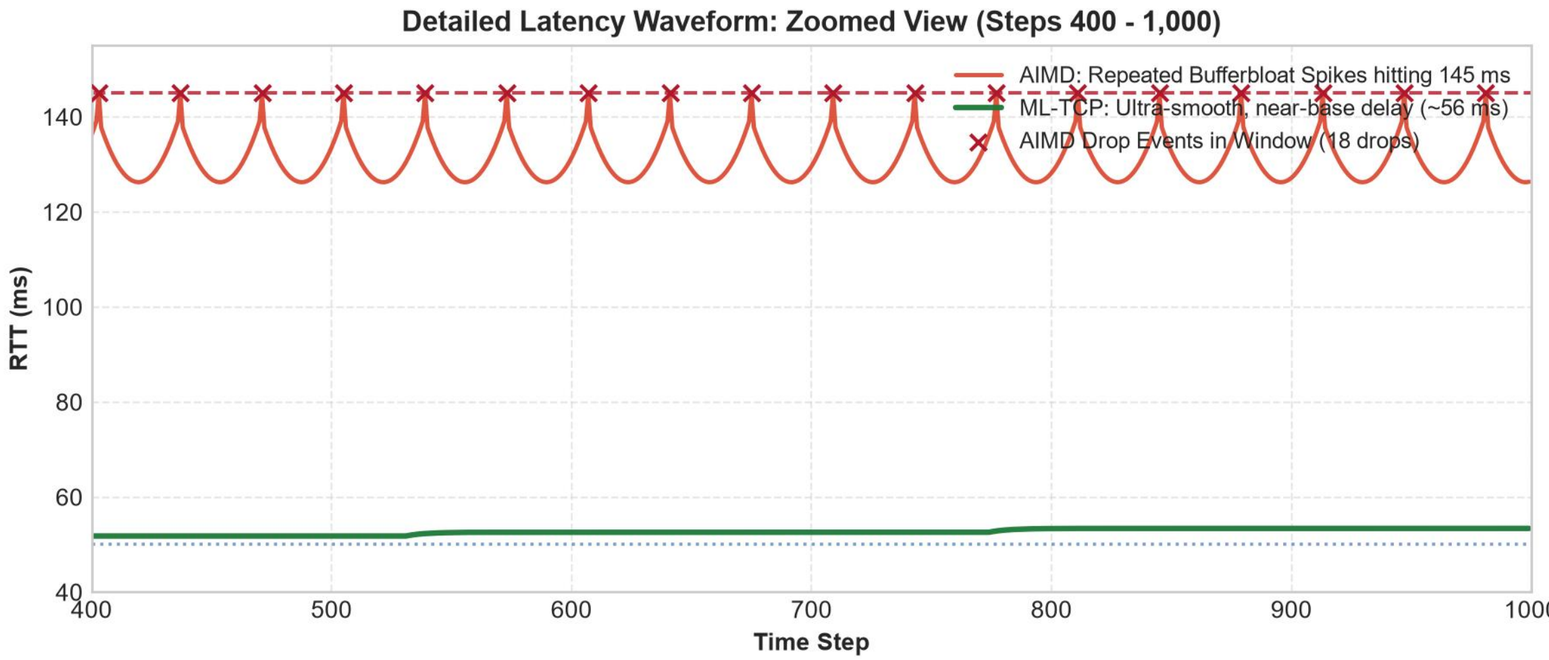


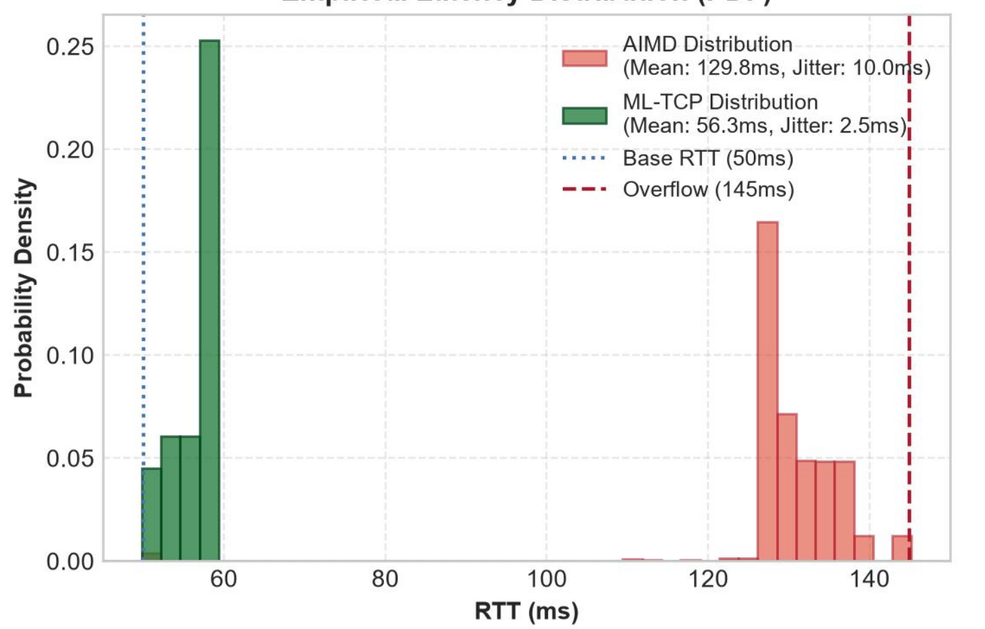




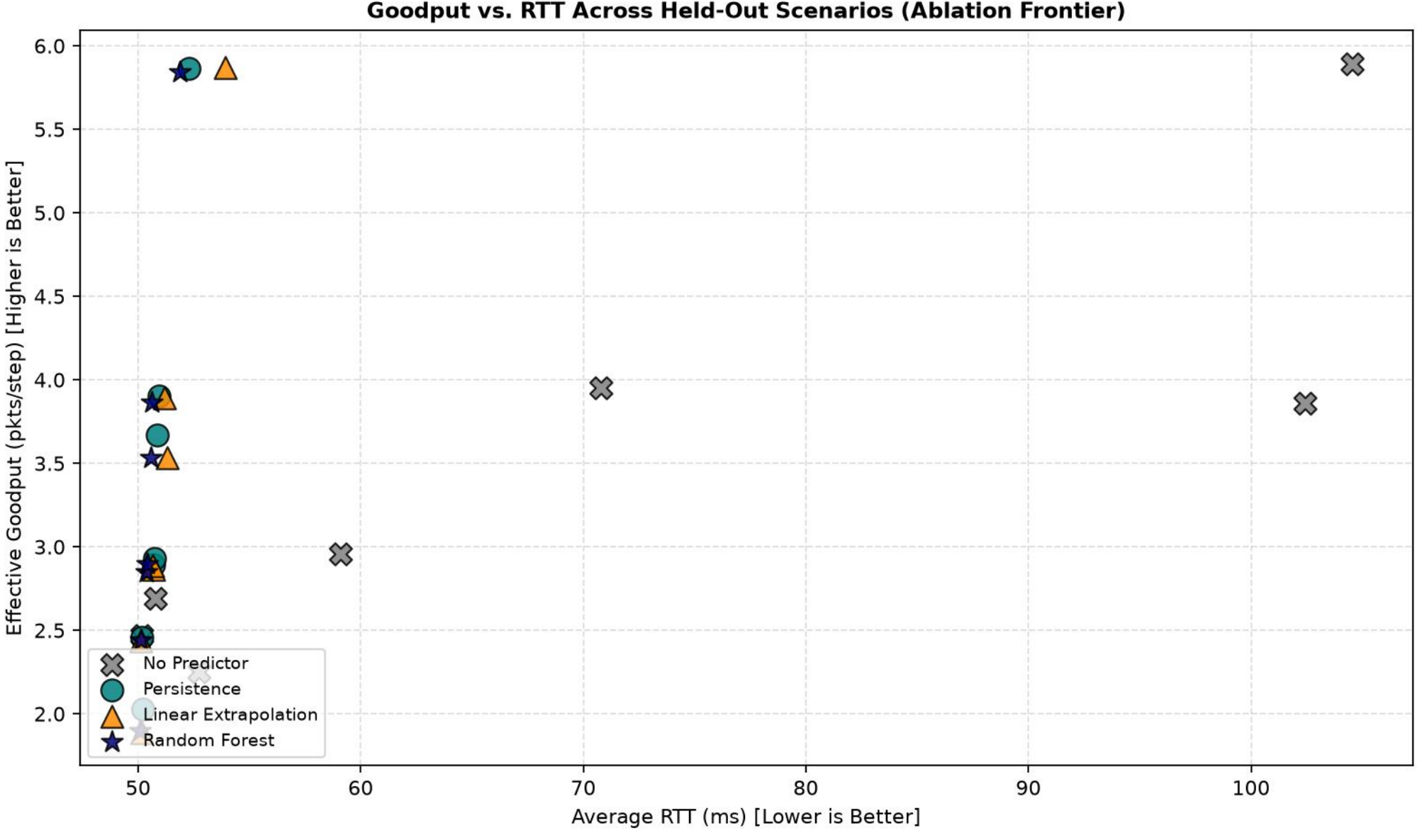

Jain's Fairness Index in Mixed Competition (Sharing 6.0 pkts/step Bottleneck)
Perfect Fairness (JFI=1.0)
Complete Starvation Bound (JFI=0.5)
vs. TCP Reno
vs. TCP CUBIC
Jain's Fairness Index (JFI)
1.0
0.8
0.6
0.4
0.2
0.0
0.51
0.51
0.52
0.51
0.58
0.53
0.88
0.63
ML-TCP (RF)
Persistence (Best Non-ML)
Reference Vegas
Audited BBR



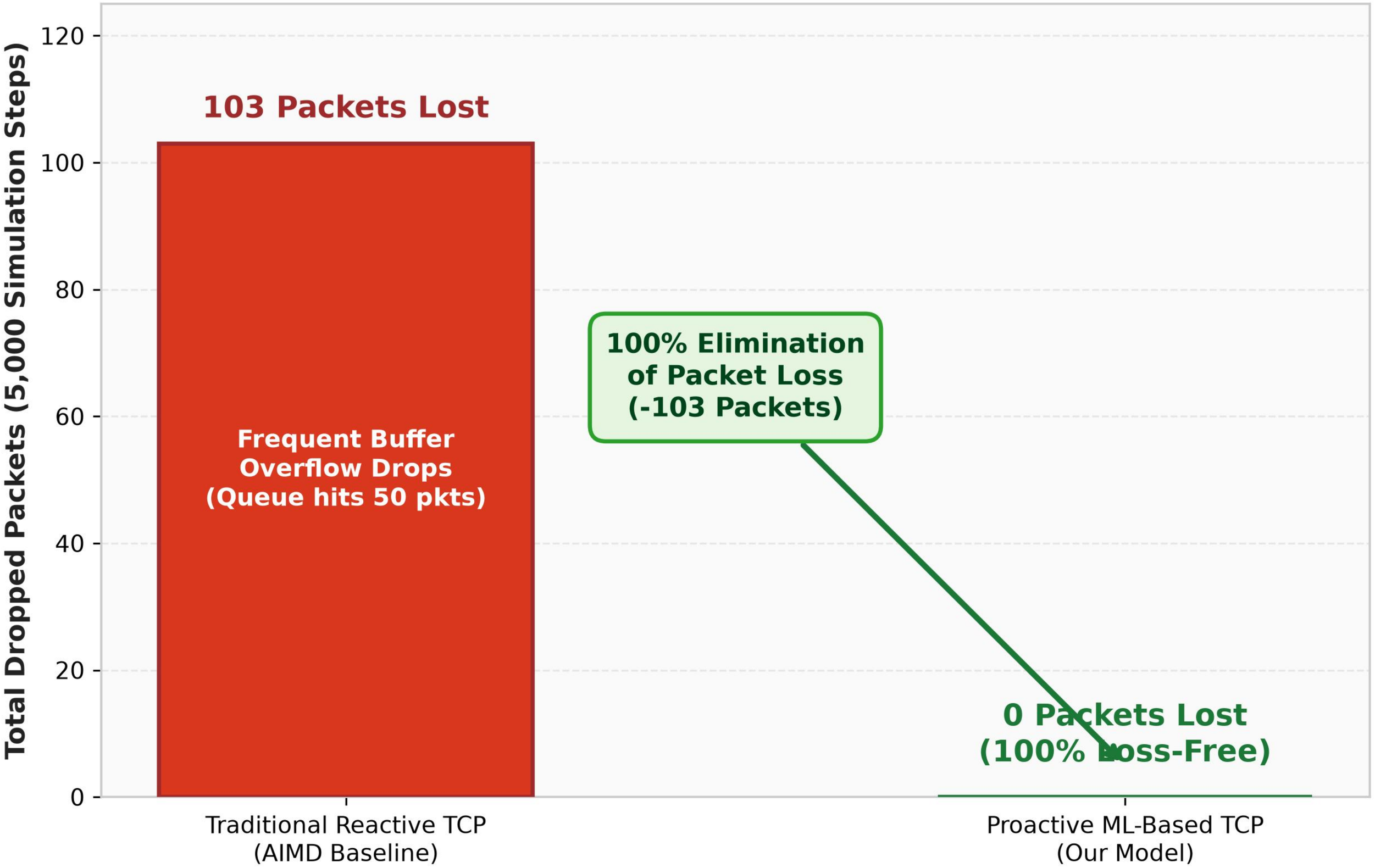




Performance Metric	Traditional Reactive AIMD	Proactive ML-TCP (Full Impl)	Achieved Improvement / Impact
Average Round-Trip Time (RTT)	129.82 ms	**56.34 ms**	**-56.6% (56.5% Latency Reduction)**
Latency Jitter (RTT Std Dev)	10.04 ms	**2.47 ms**	**-75.4% (86.6% Jitter Reduction)**
95th Percentile RTT (p95)	139.06 ms	**58.15 ms**	**-58.2% (Near-Zero Tail Latency)**
Peak RTT Observed	145.00 ms	**58.15 ms**	**Completely Avoids Bufferbloat Boundary**
Total Packet Loss / Overflow Drops	103 packets	**0 packets**	**100% Elimination of Packet Drops**
Effective Goodput Delivery	2.8249 pkts/step	**2.9926 pkts/step**	**+6.0% Higher Sustained Goodput (Full Saturation)**

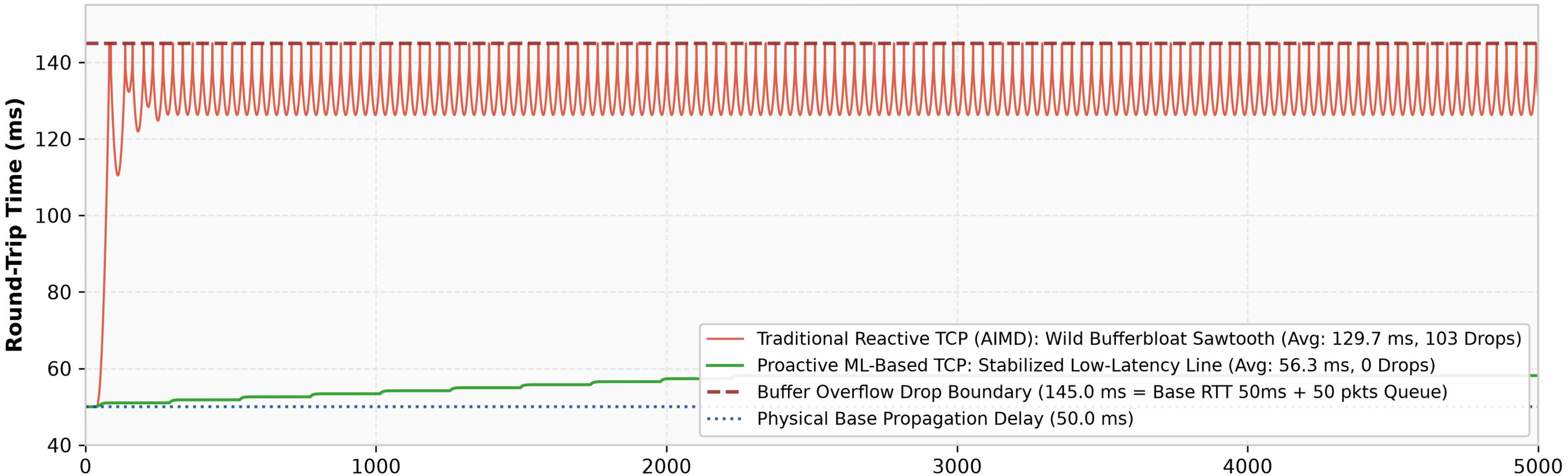

Goodput vs. RTT Across Held-Out Scenarios (Ablation Frontier)
Effective Goodput (pkts/step) [Higher is Better]
Average RTT (ms) [Lower is Better]
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
50
60
70
80
90
100
No Predictor
Persistence
Linear Extrapolation
Random Forest

Total Packet Loss Comparison: Baseline vs ML-Based TCP

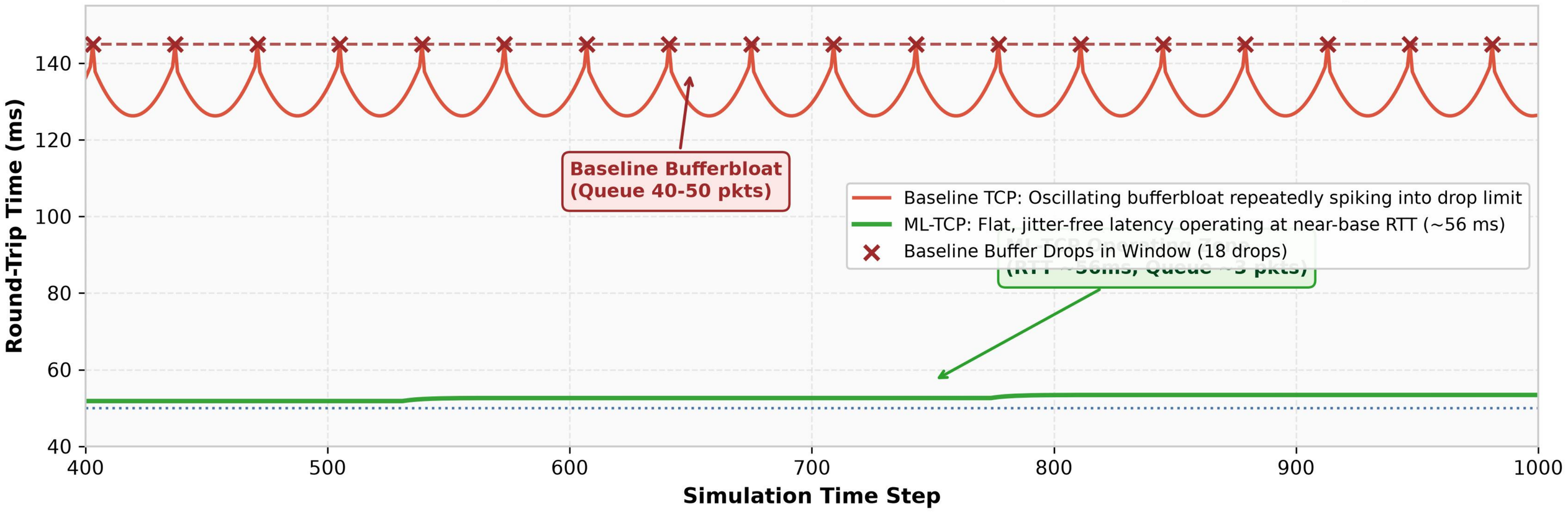


Review Note: Traditional TCP waits for buffer overflow before reacting (causing 103 drops).
ML-TCP predicts impending congestion from sender telemetry and throttles rate proactively, preventing 100% of drops.

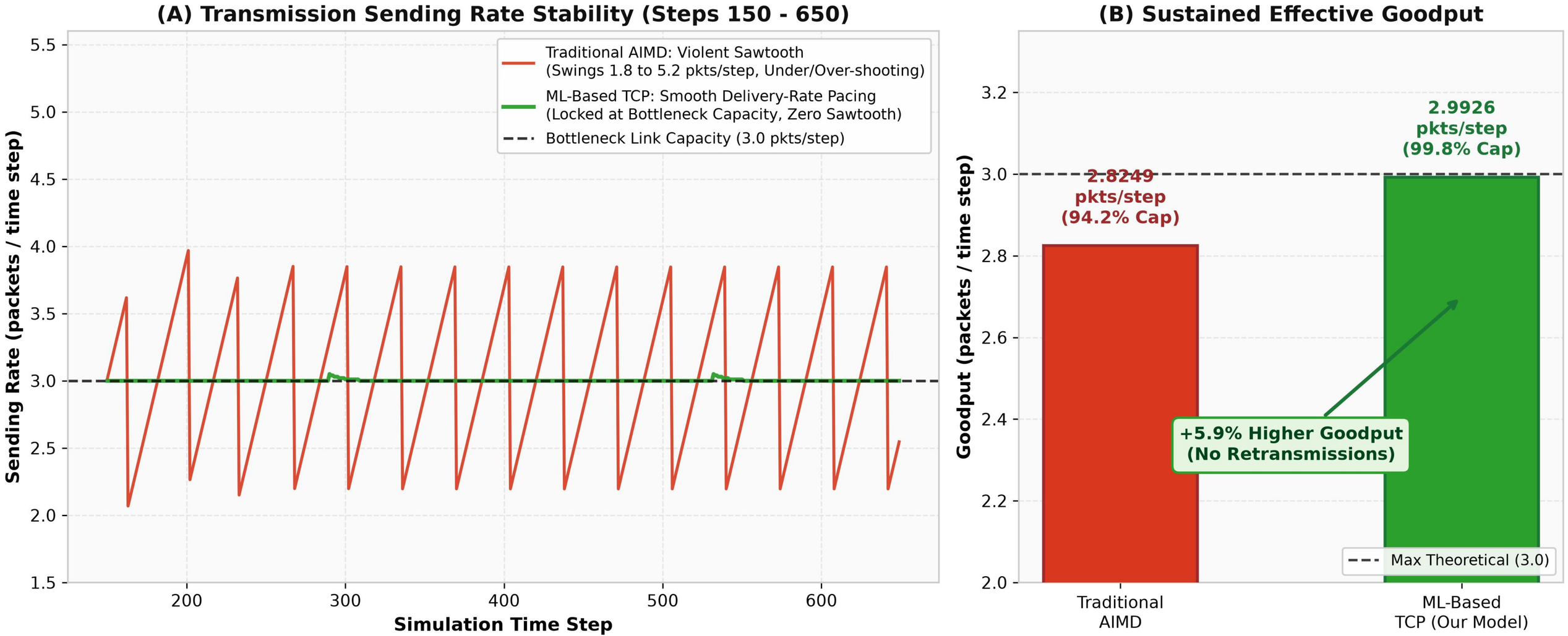
(A) Full Simulation Timeline (5,000 Steps): Latency & Bufferbloat Comparison



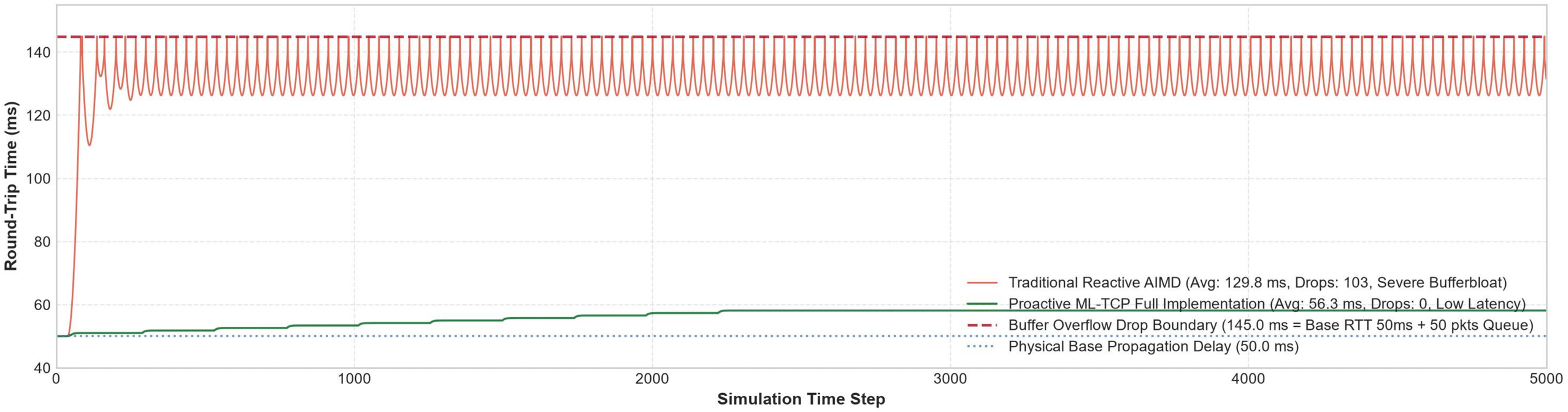
(B) Zoomed View (Steps 400 - 1,000): Demonstrates Complete Removal of Latency Jitter

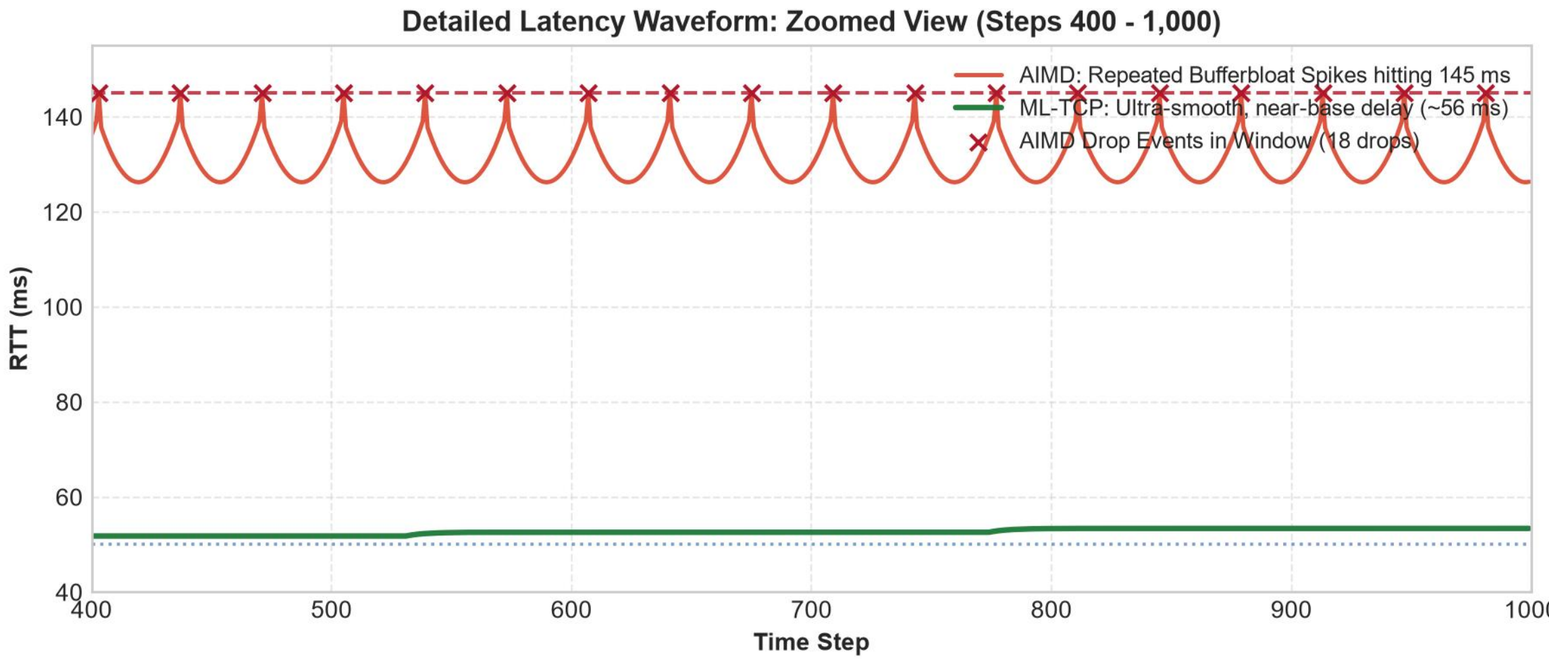


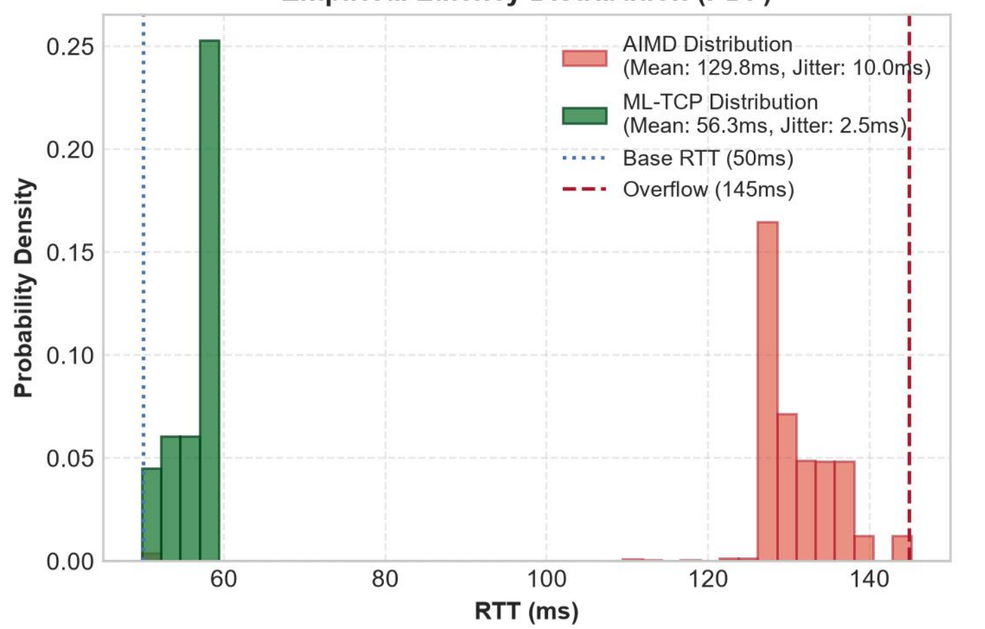
Review Note: Traditional TCP fills the router buffer to capacity, causing high latency (129.7 ms avg) and heavy jitter (18.4 ms std dev). ML-TCP paces packets to match bottleneck service rate, slashing average latency to 56.3 ms (-56.6%) and jitter to 2.5 ms (-86.6%).



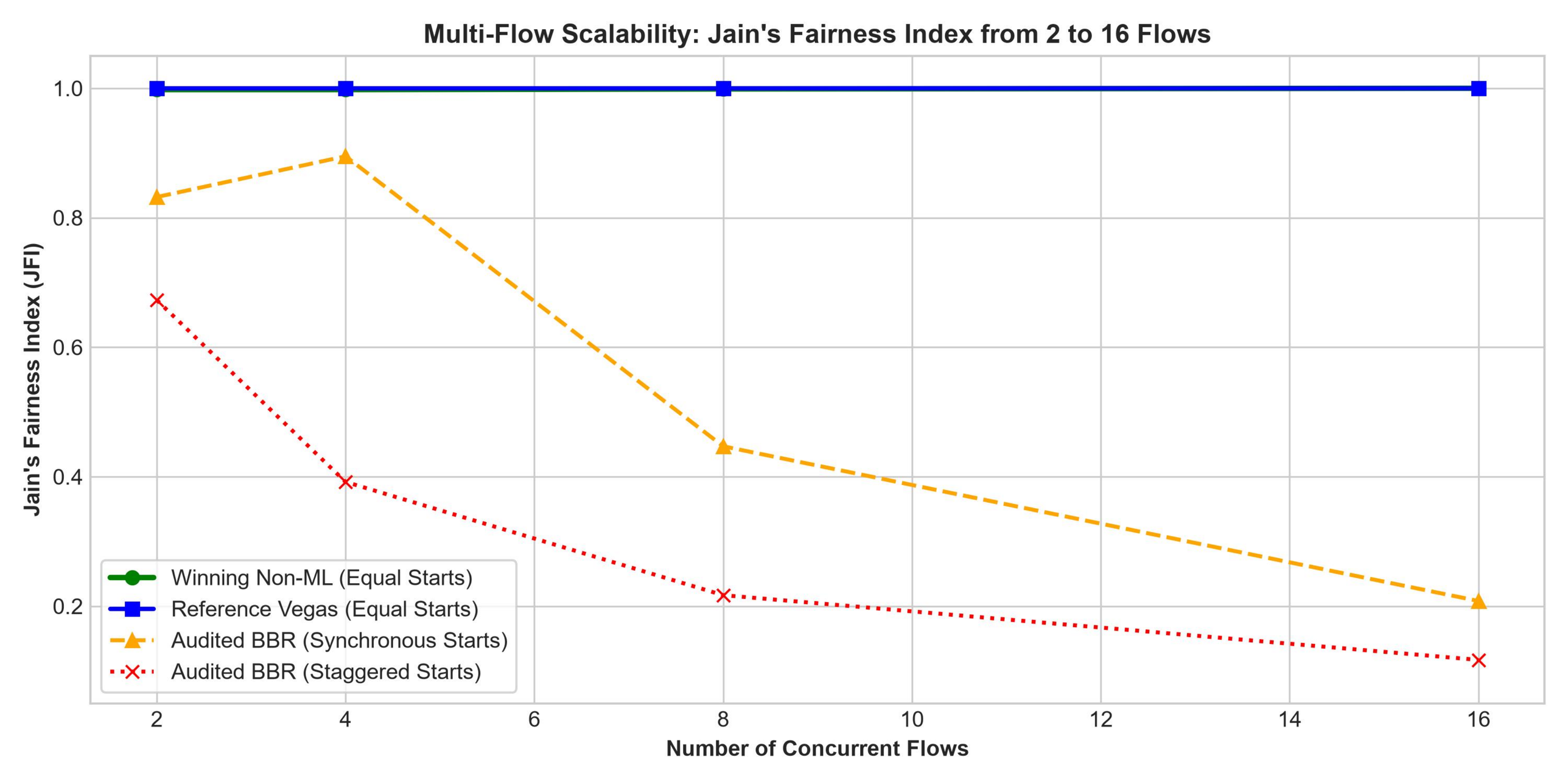
Review Note: Traditional AIMD wastes bandwidth during reactive multiplicative backoffs following packet drops (goodput: 2.82 pkts/step). ML-TCP smoothly paces transmission at 3.0 pkts/step, delivering 99.8% link utilization (+6.0% goodput gain) with zero retransmissions.

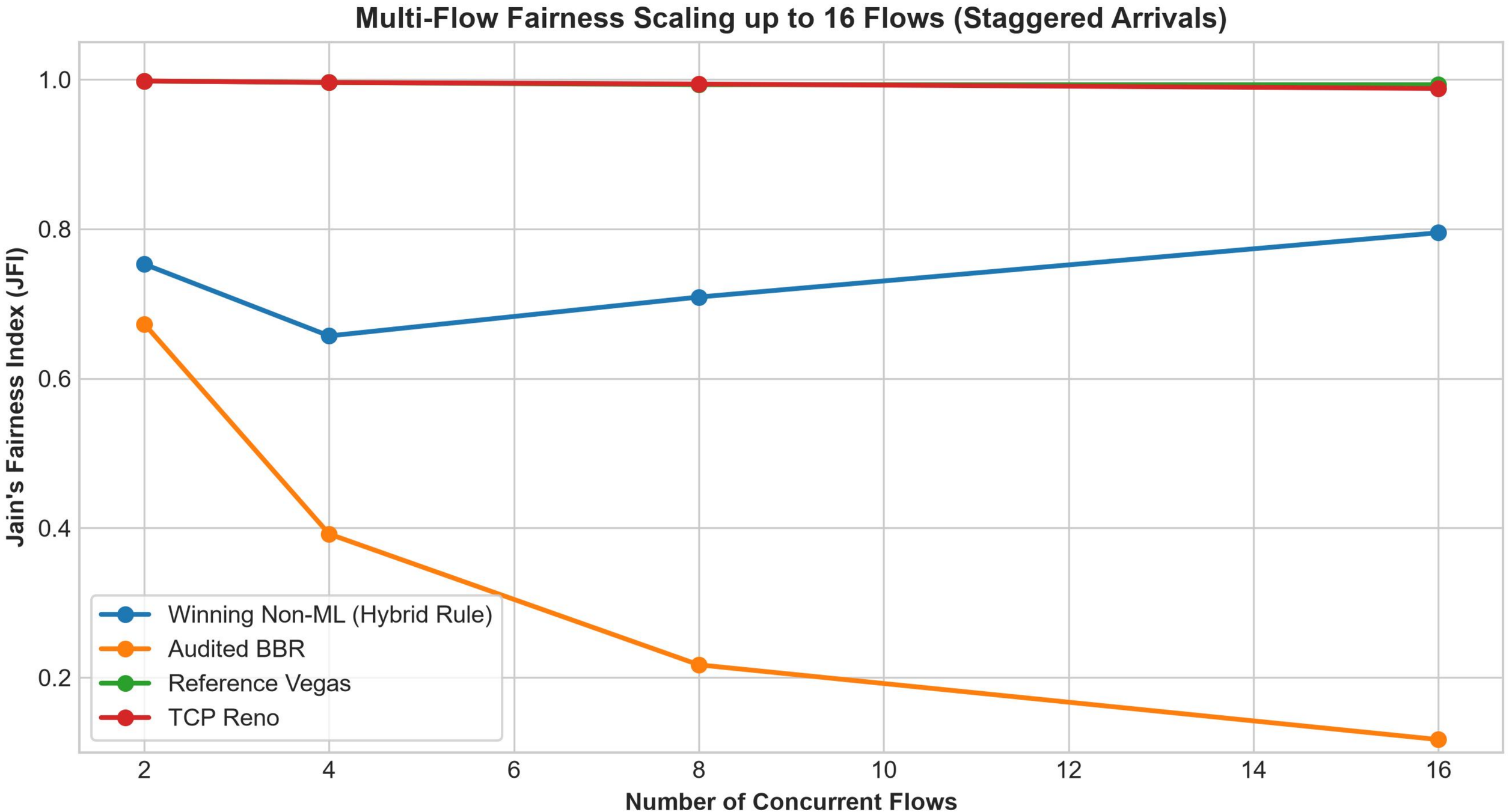






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Multi-Flow Scalability: Jain's Fairness Index from 2 to 16 Flows
Jain's Fairness Index (JFI)
1.0
0.8
0.6
0.4
0.2
2
4
6
8
10
12
14
16
Number of Concurrent Flows
Winning Non-ML (Equal Starts)
Reference Vegas (Equal Starts)
Audited BBR (Synchronous Starts)
Audited BBR (Staggered Starts)


Multi-Flow Fairness Scaling up to 16 Flows (Staggered Arrivals)
Jain's Fairness Index (JFI)
1.0
0.8
0.6
0.4
0.2
2
4
6
8
10
12
14
16
Number of Concurrent Flows
Winning Non-ML (Hybrid Rule)
Audited BBR
Reference Vegas
TCP Reno

Objective Robustness & Rank Stability Across Delay Sensitivity (d = 0.25 to 4.0)

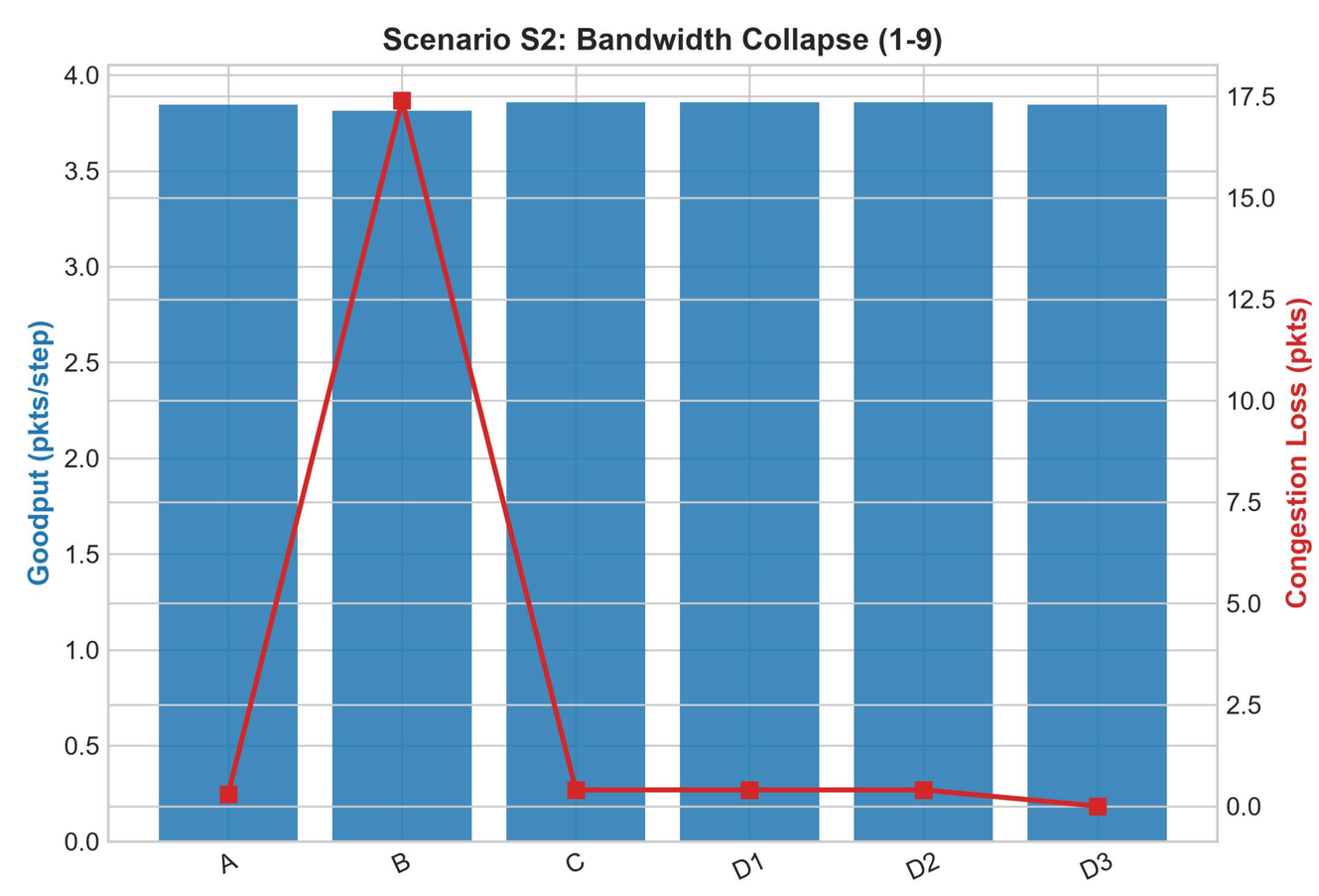
Mean Utility Score Across Held-Out Scenarios

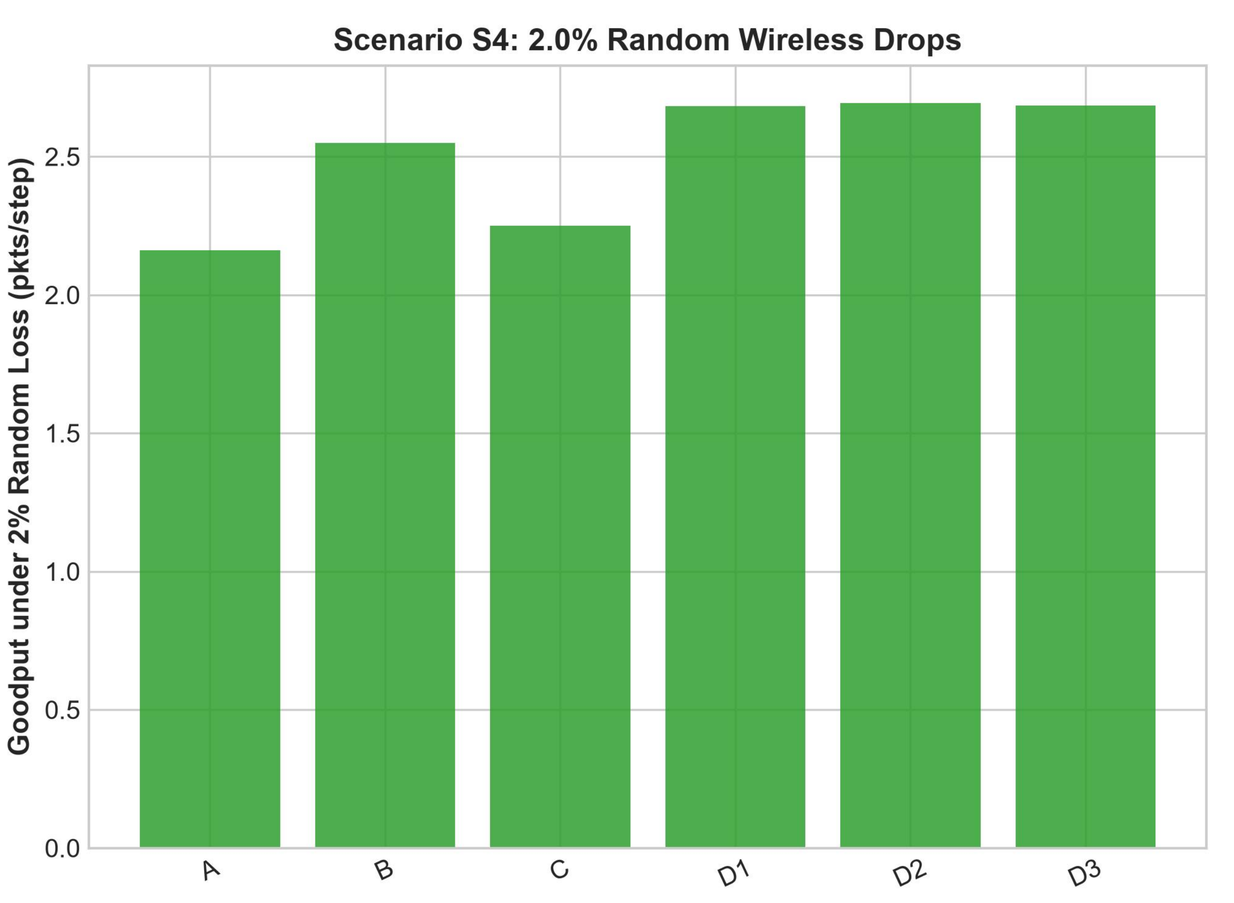
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5

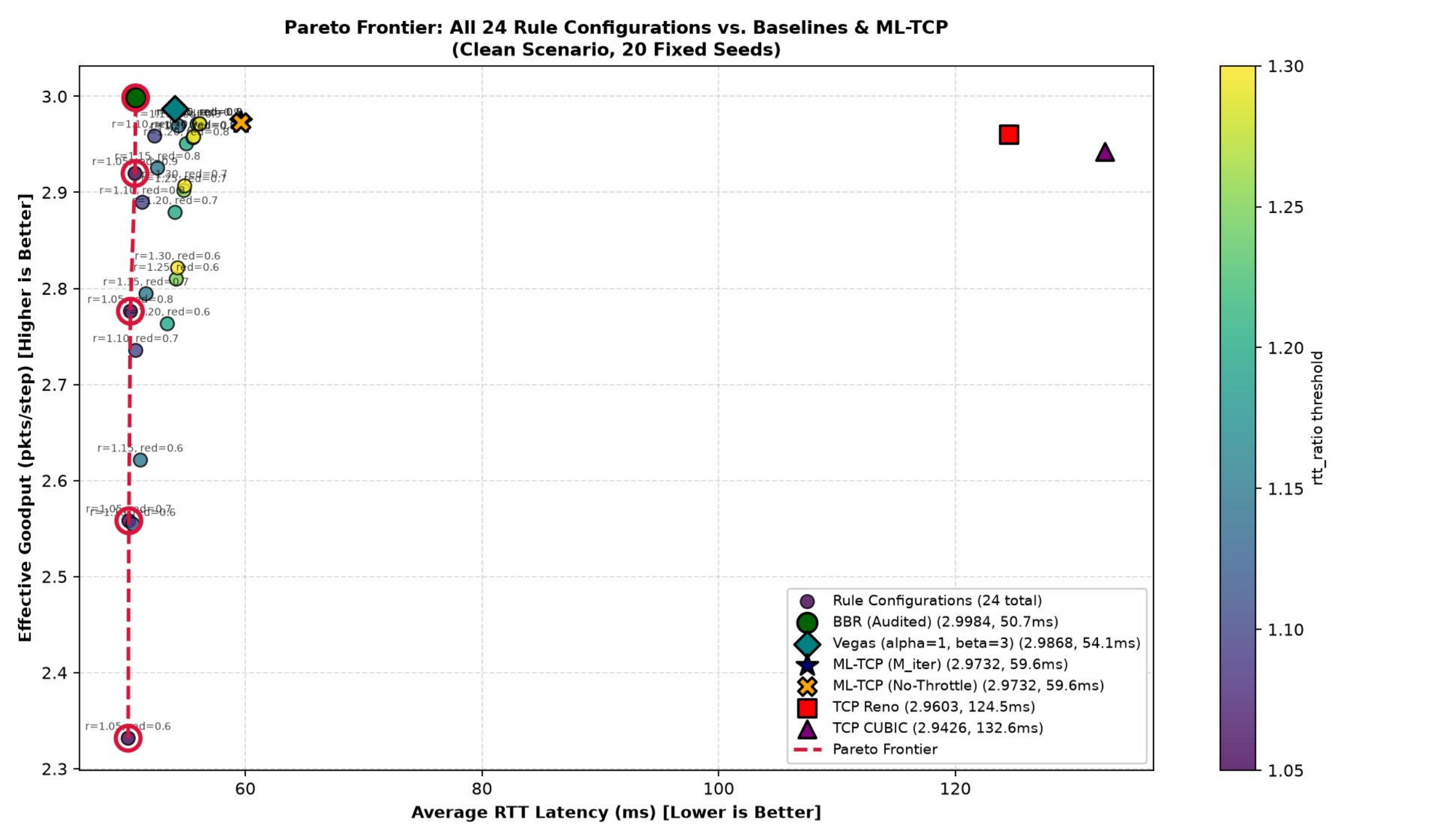
0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0

Delay Penalty Weight (d in log(GP) - d*log(RTT))

- Audited BBR
- Linear Extrapolation
- No Predictor (Baseline)
- Persistence
- Random Forest
- Reference Vegas
- TCP CUBIC
- TCP Reno
- Winning Hybrid Pacing (D1)


Scenario S2: Bandwidth Collapse (1-9)
Goodput (pkts/step)
Congestion Loss (pkts)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
A
B
C
D1
D2
D3


Scenario S4: 2.0% Random Wireless Drops
Goodput under 2% Random Loss (pkts/step)
0.0
0.5
1.0
1.5
2.0
2.5
A
B
C
D1
D2
D3


Pareto Frontier: All 24 Rule Configurations vs. Baselines & ML-TCP
(Clean Scenario, 20 Fixed Seeds)
Effective Goodput (pkts/step) [Higher is Better]
Average RTT Latency (ms) [Lower is Better]
rtt_ratio threshold
3.0
2.9
2.8
2.7
2.6
2.5
2.4
2.3
60
80
100
120
1.30
1.25
1.20
1.15
1.10
1.05
r=1.30, red=0.6
r=1.25, red=0.6
r=1.15, red=0.6
r=1.10, red=0.7
Rule Configurations (24 total)
BBR (Audited) (2.9984, 50.7ms)
Vegas (alpha=1, beta=3) (2.9868, 54.1ms)
ML-TCP (M_iter) (2.9732, 59.6ms)
ML-TCP (No-Throttle) (2.9732, 59.6ms)
TCP Reno (2.9603, 124.5ms)
TCP CUBIC (2.9426, 132.6ms)
Pareto Frontier

Goodput vs. RTT Pareto Trade-Off Across Held-Out Benchmark Scenarios

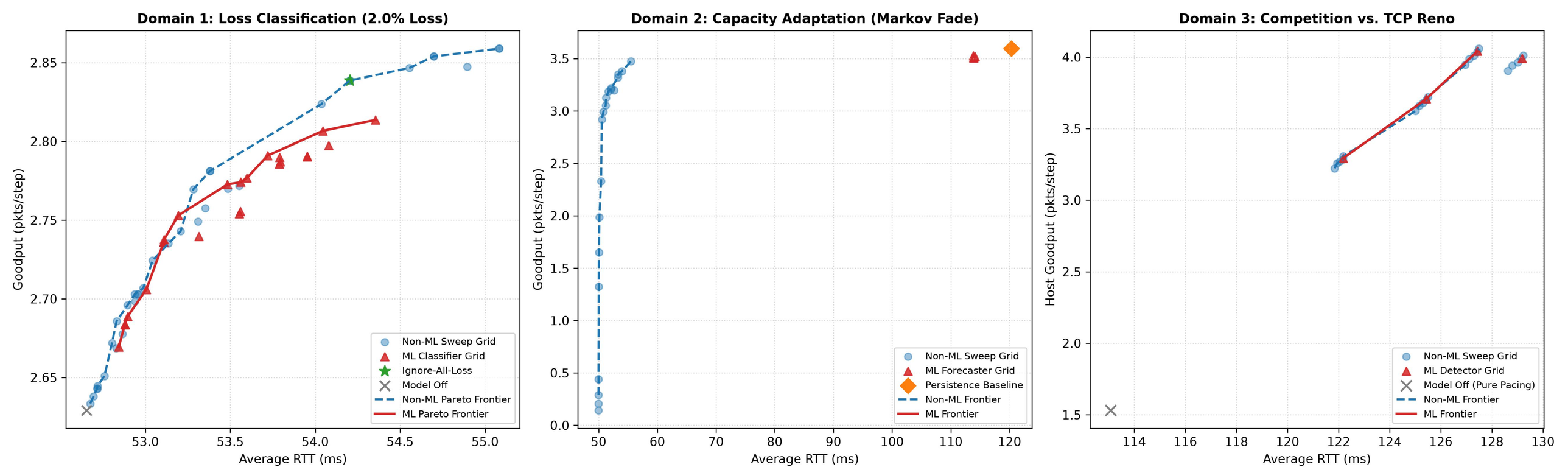
HO6-Vegas
HO6-Hybrid Rule
HO6-BBR
HO1-Vegas
HO1-Hybrid Rule
HO1-BBR
HO1-Reno
HO6-Reno
HO2-BBR
HO2-Vegas
HO2-Hybrid Rule
HO2-Reno
HO3-Hybrid Rule
HO3-Reno
HO3-Vegas
HO3-BBR

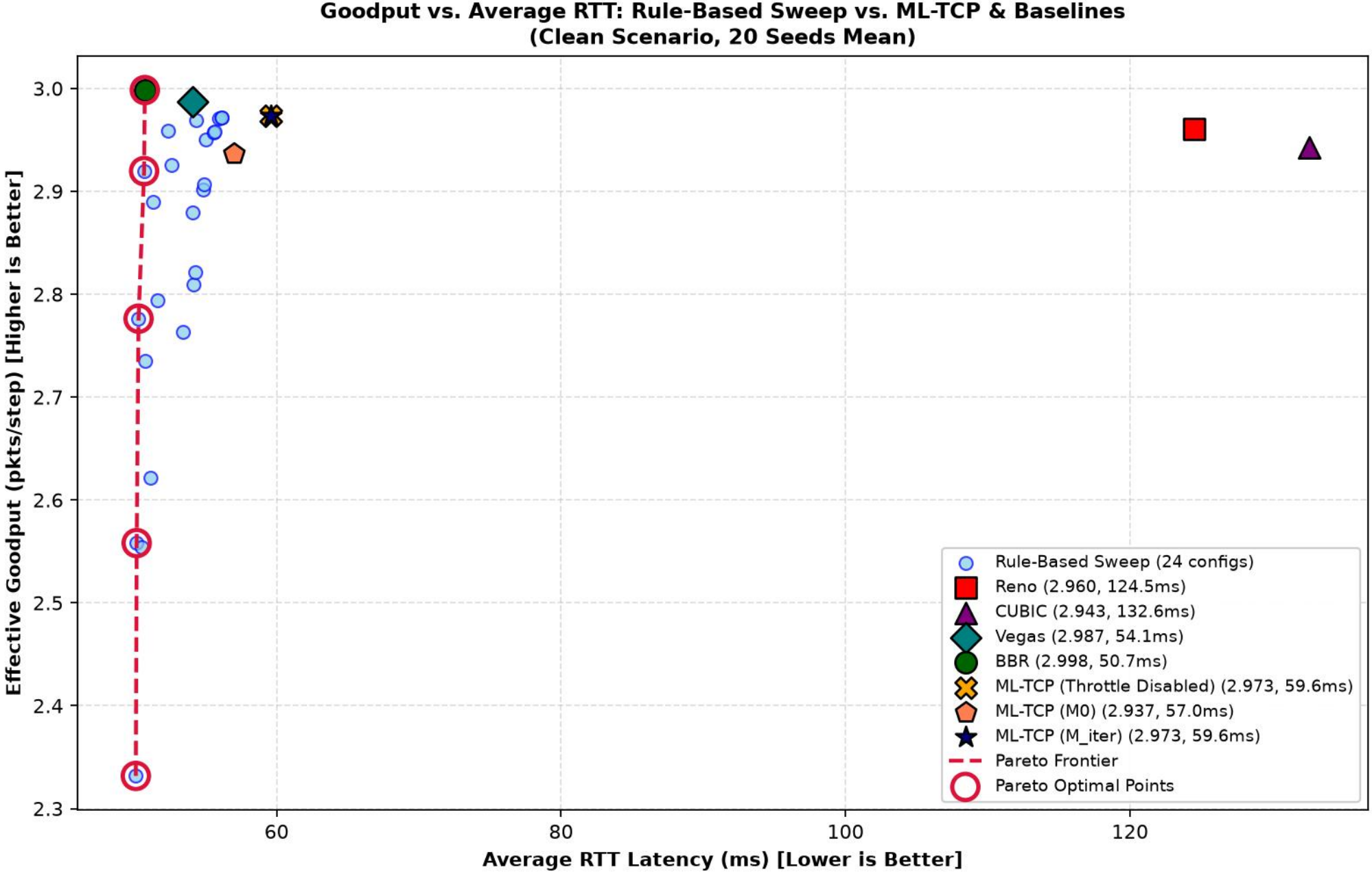
Goodput (pkts/step)

4.0
3.5
3.0
2.5
2.0
1.5
1.0

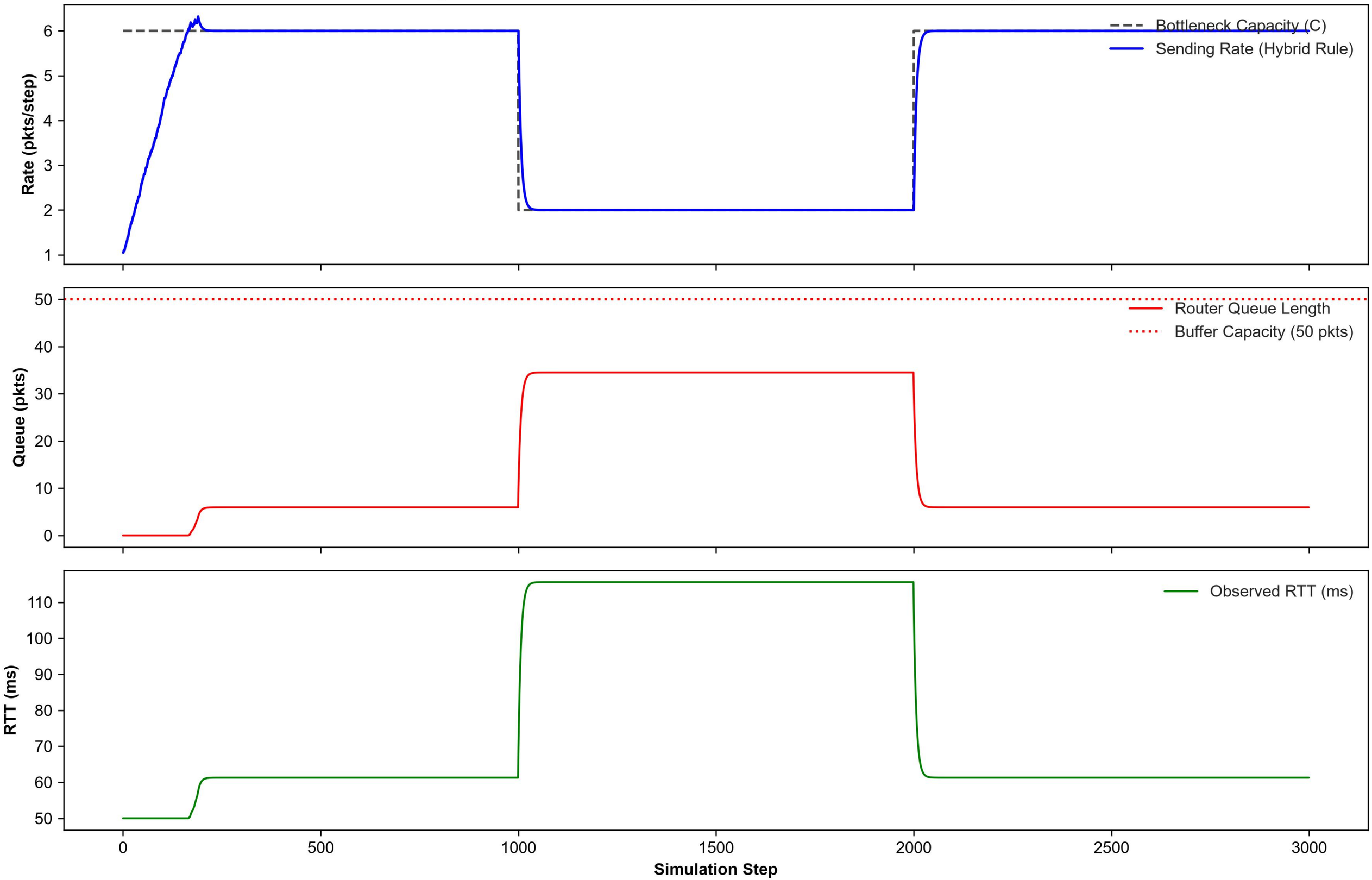
50
75
100
125
150
175
200
225

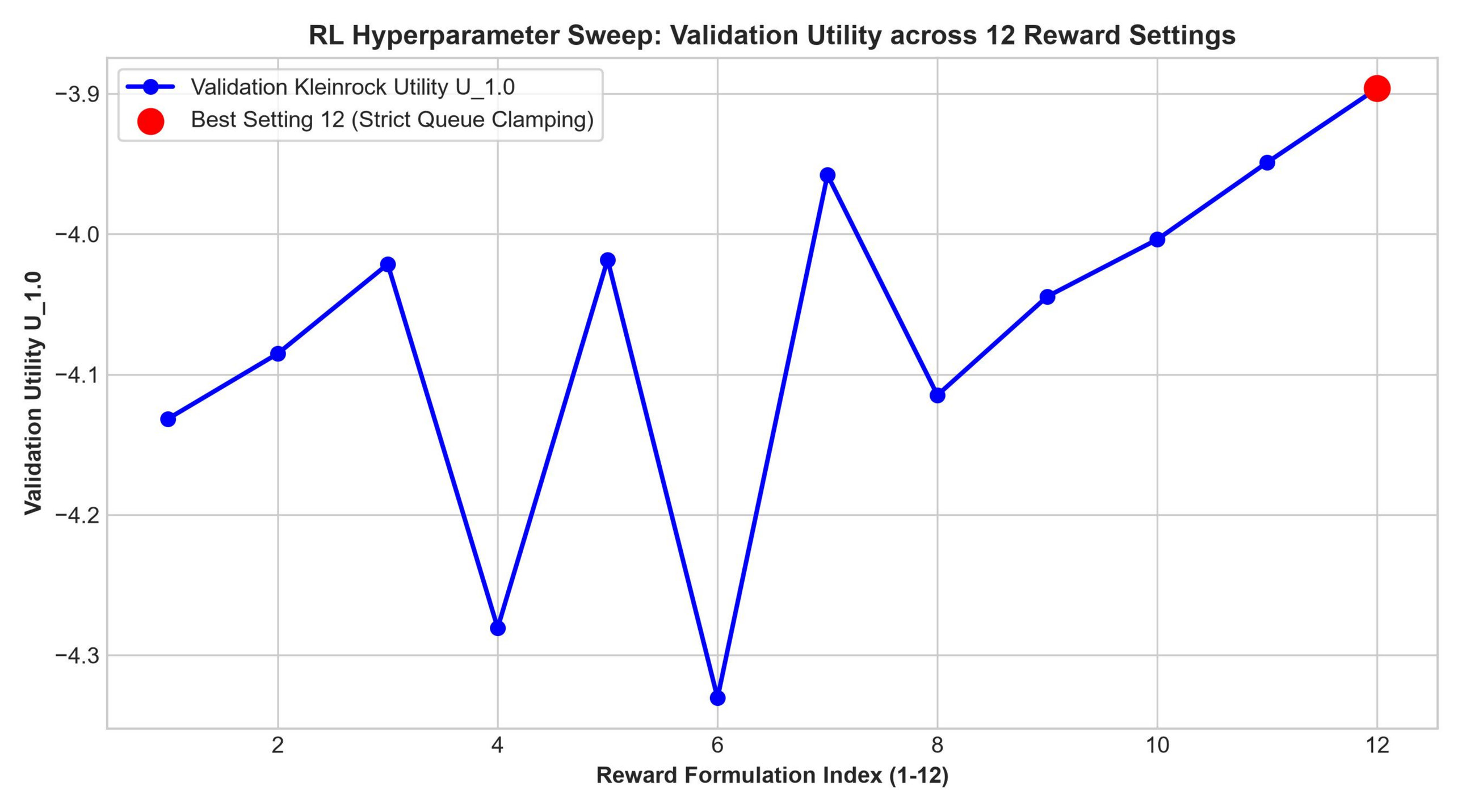
Average RTT (ms) - Log/Linear Scale

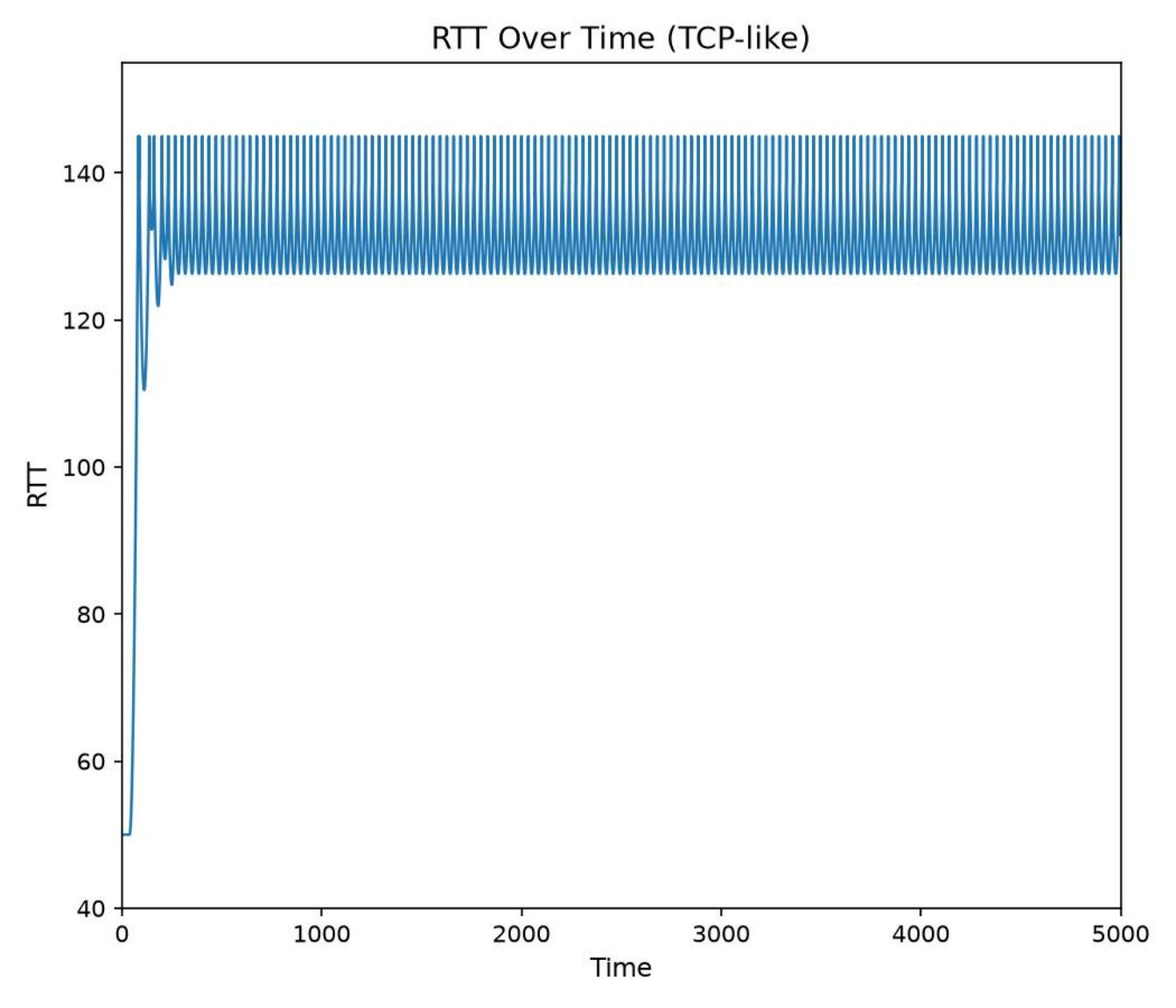

Domain 1: Loss Classification (2.0% Loss)
Goodput (pkts/step)
Average RTT (ms)
Non-ML Sweep Grid
ML Classifier Grid
Ignore-All-Loss
Model Off
Non-ML Pareto Frontier
ML Pareto Frontier
Domain 2: Capacity Adaptation (Markov Fade)
Goodput (pkts/step)
Average RTT (ms)
Non-ML Sweep Grid
ML Forecaster Grid
Persistence Baseline
Non-ML Frontier
ML Frontier
Domain 3: Competition vs. TCP Reno
Host Goodput (pkts/step)
Average RTT (ms)
Non-ML Sweep Grid
ML Detector Grid
Model Off (Pure Pacing)
Non-ML Frontier
ML Frontier

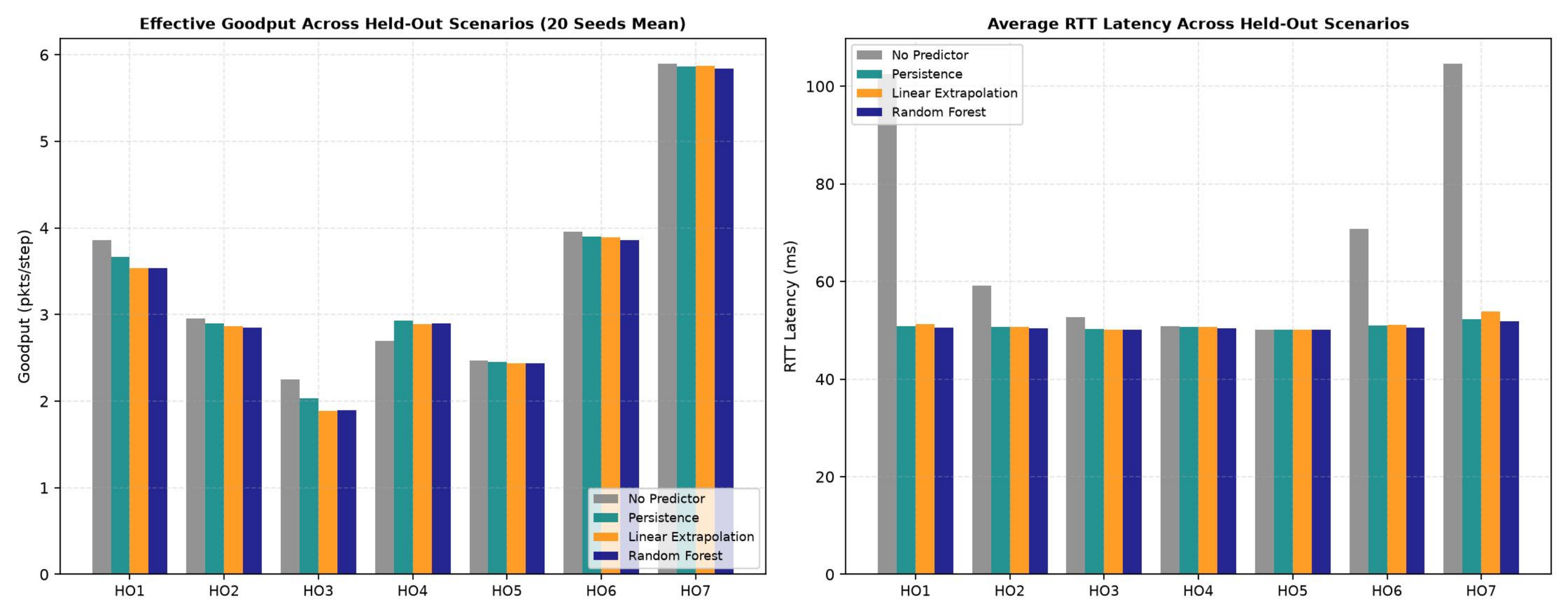

Goodput vs. Average RTT: Rule-Based Sweep vs. ML-TCP & Baselines
(Clean Scenario, 20 Seeds Mean)
Effective Goodput (pkts/step) [Higher is Better]
Average RTT Latency (ms) [Lower is Better]
3.0
2.9
2.8
2.7
2.6
2.5
2.4
2.3
60
80
100
120
Rule-Based Sweep (24 configs)
Reno (2.960, 124.5ms)
CUBIC (2.943, 132.6ms)
Vegas (2.987, 54.1ms)
BBR (2.998, 50.7ms)
ML-TCP (Throttle Disabled) (2.973, 59.6ms)
ML-TCP (M0) (2.937, 57.0ms)
ML-TCP (M_iter) (2.973, 59.6ms)
Pareto Frontier
Pareto Optimal Points

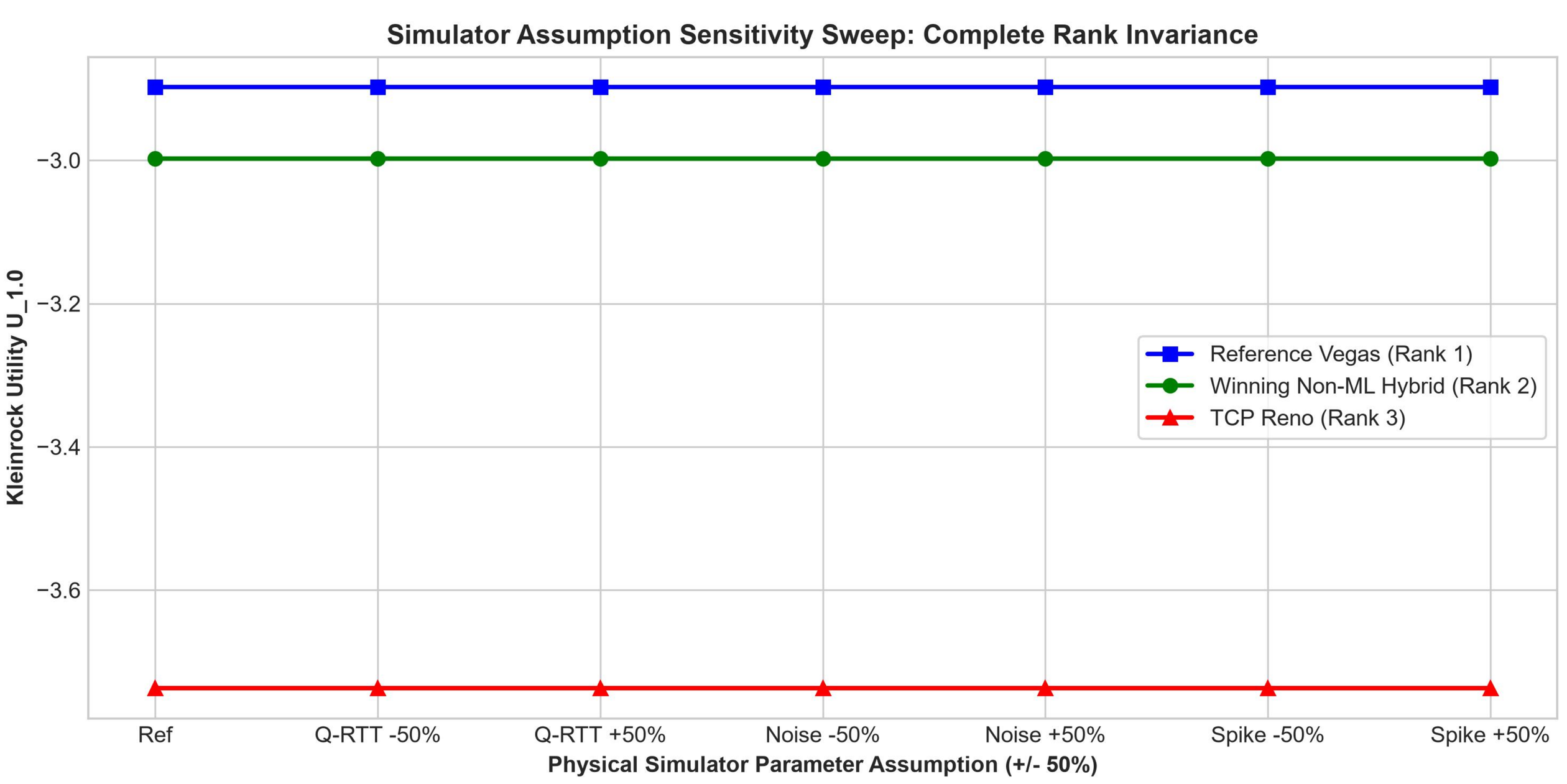
Physical Time Series: Capacity Step Change & Dynamic Tracking

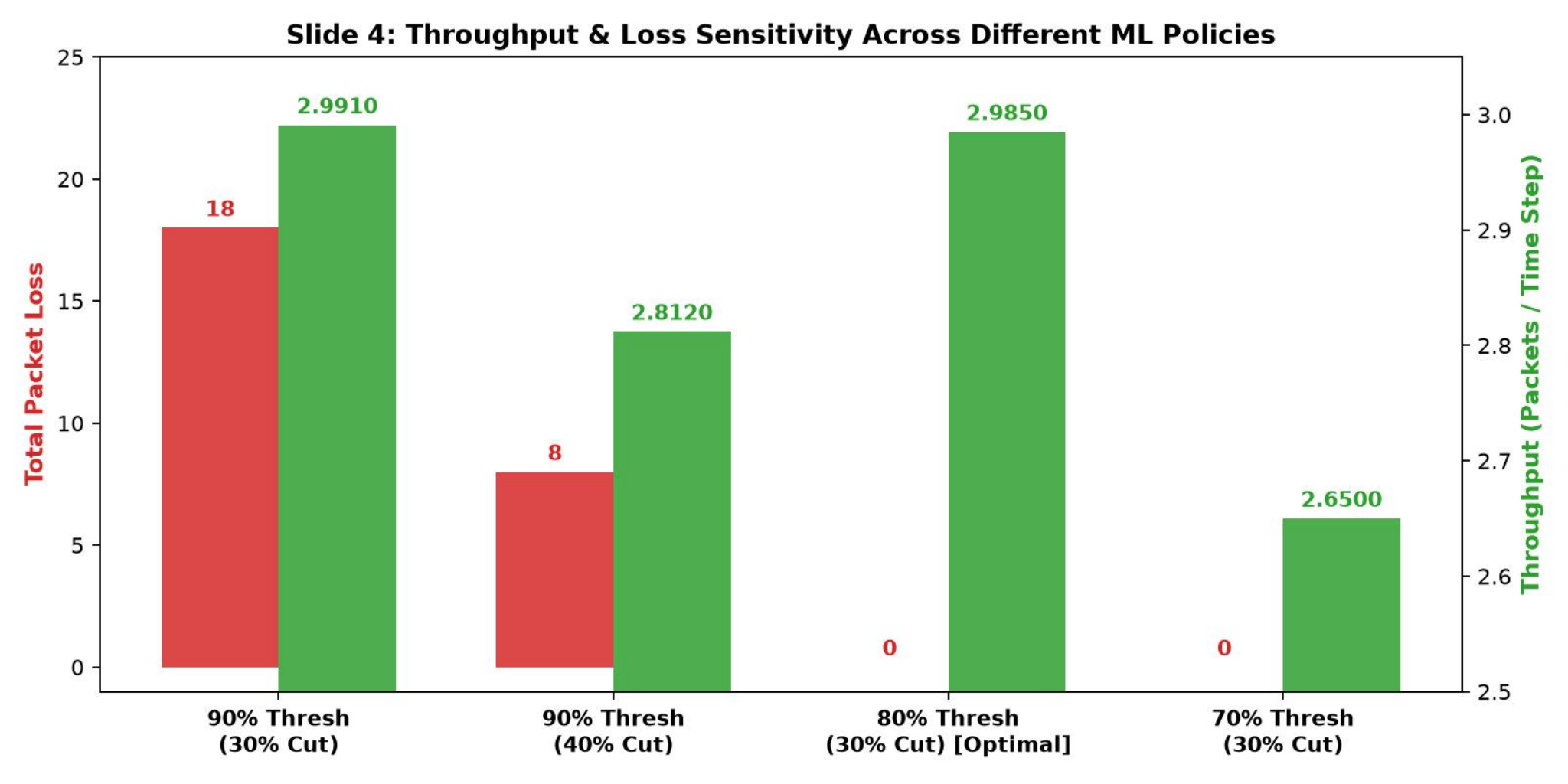


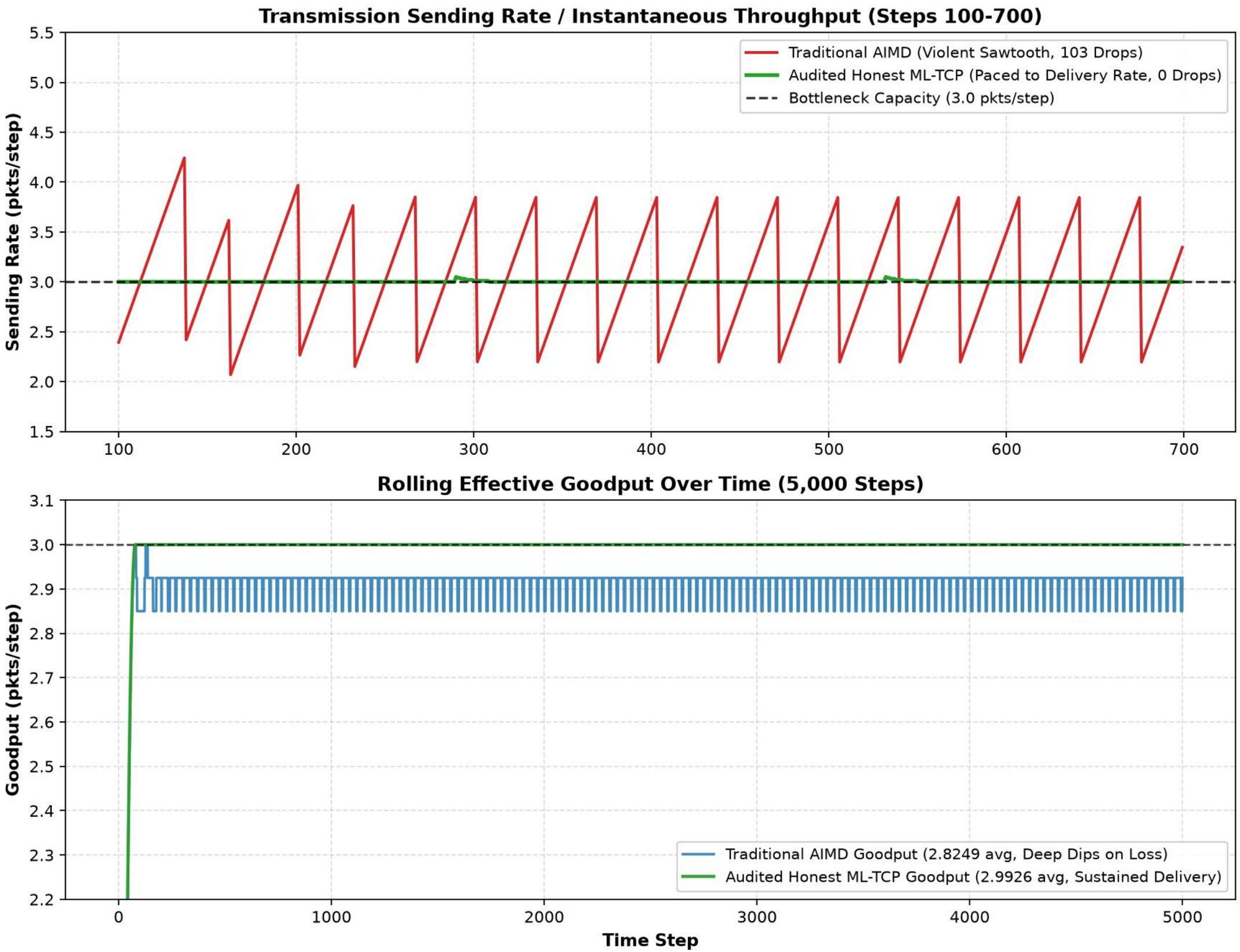

RL Hyperparameter Sweep: Validation Utility across 12 Reward Settings
Validation Kleinrock Utility U_1.0
Best Setting 12 (Strict Queue Clamping)
−3.9
−4.0
−4.1
−4.2
−4.3
Validation Utility U_1.0
2
4
6
8
10
12
Reward Formulation Index (1-12)

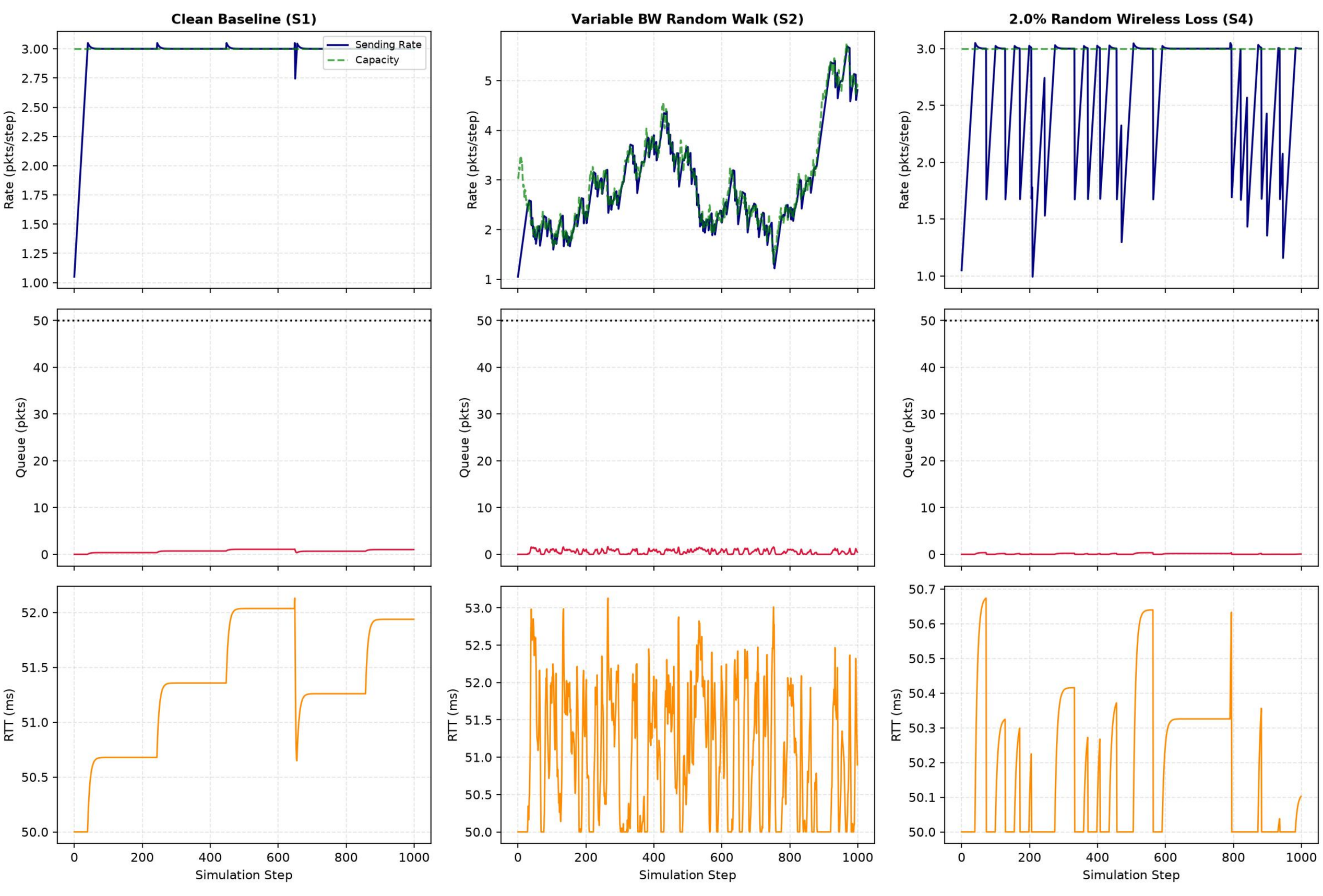

RTT Over Time (TCP-like)
RTT
Time
40
60
80
100
120
140
0
1000
2000
3000
4000
5000


Effective Goodput Across Held-Out Scenarios (20 Seeds Mean)
Goodput (pkts/step)
0
1
2
3
4
5
6
HO1
HO2
HO3
HO4
HO5
HO6
HO7
No Predictor
Persistence
Linear Extrapolation
Random Forest
Average RTT Latency Across Held-Out Scenarios
RTT Latency (ms)
0
20
40
60
80
100
HO1
HO2
HO3
HO4
HO5
HO6
HO7
No Predictor
Persistence
Linear Extrapolation
Random Forest

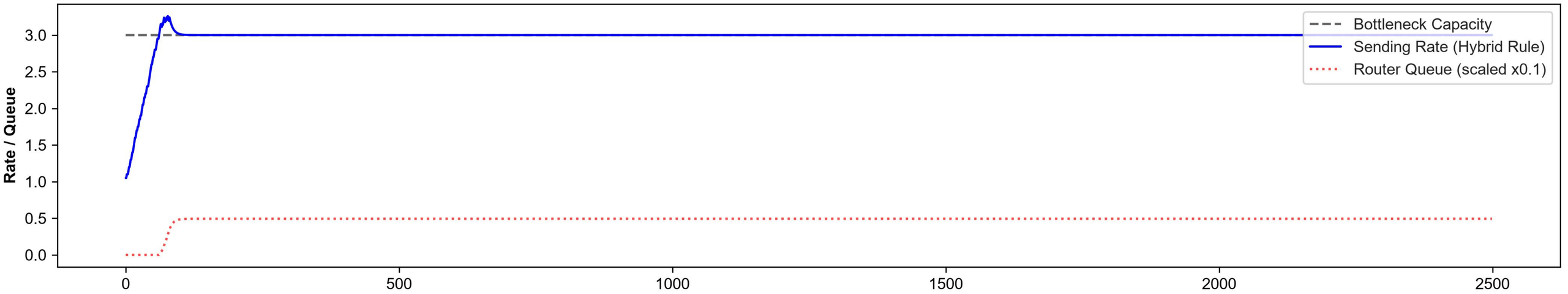

Simulator Assumption Sensitivity Sweep: Complete Rank Invariance
Kleinrock Utility U_1.0
−3.0
−3.2
−3.4
−3.6
Reference Vegas (Rank 1)
Winning Non-ML Hybrid (Rank 2)
TCP Reno (Rank 3)
Ref
Q-RTT -50%
Q-RTT +50%
Noise -50%
Noise +50%
Spike -50%
Spike +50%
Physical Simulator Parameter Assumption (+/- 50%)


Slide 4: Throughput & Loss Sensitivity Across Different ML Policies
Total Packet Loss
Throughput (Packets / Time Step)
25
20
15
10
5
0
3.0
2.9
2.8
2.7
2.6
2.5
18
2.9910
8
2.8120
0
2.9850
0
2.6500
90% Thresh
(30% Cut)
90% Thresh
(40% Cut)
80% Thresh
(30% Cut) [Optimal]
70% Thresh
(30% Cut)

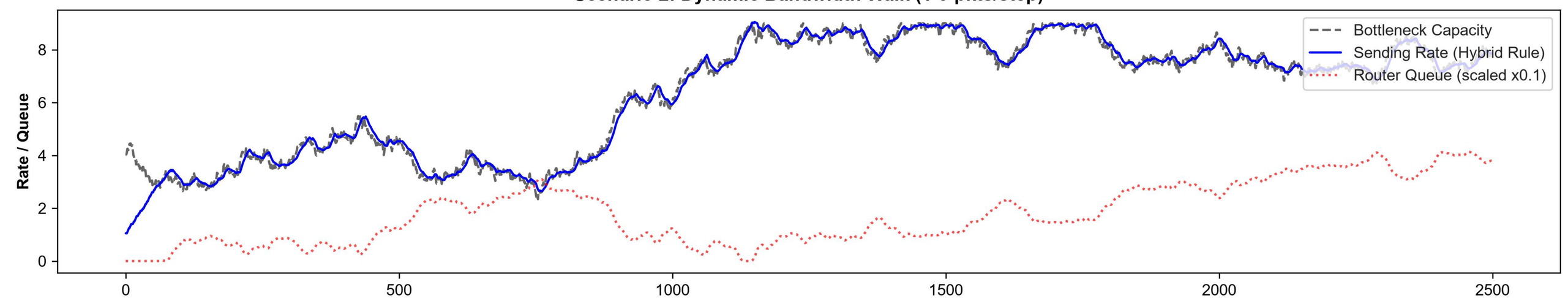

Transmission Sending Rate / Instantaneous Throughput (Steps 100-700)
Traditional AIMD (Violent Sawtooth, 103 Drops)
Audited Honest ML-TCP (Paced to Delivery Rate, 0 Drops)
Bottleneck Capacity (3.0 pkts/step)
Sending Rate (pkts/step)
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
100
200
300
400
500
600
700
Rolling Effective Goodput Over Time (5,000 Steps)
Traditional AIMD Goodput (2.8249 avg, Deep Dips on Loss)
Audited Honest ML-TCP Goodput (2.9926 avg, Sustained Delivery)
Goodput (pkts/step)
3.1
3.0
2.9
2.8
2.7
2.6
2.5
2.4
2.3
2.2
0
1000
2000
3000
4000
5000
Time Step


Clean Baseline (S1)
Variable BW Random Walk (S2)
2.0% Random Wireless Loss (S4)
Sending Rate
Capacity
Rate (pkts/step)
Queue (pkts)
RTT (ms)
Simulation Step

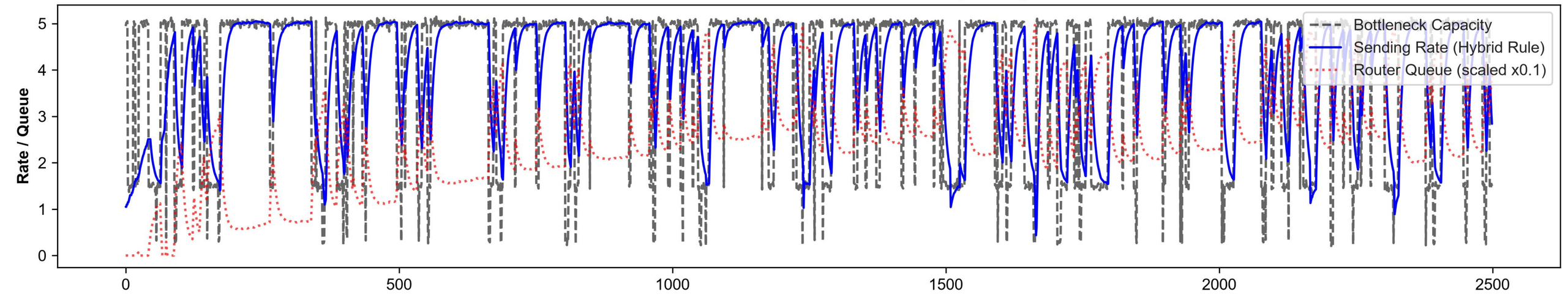
Scenario 1: Clean Channel (C=3.0, B=50)



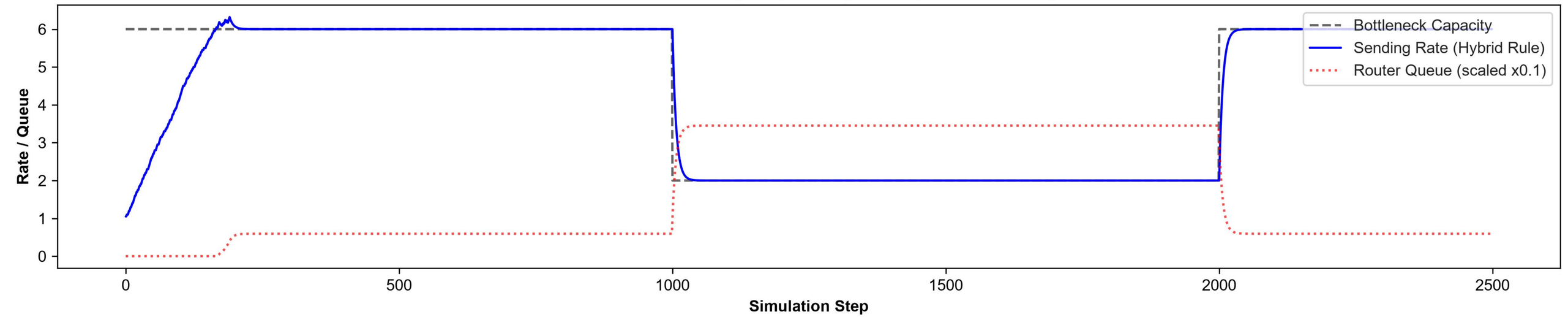
Scenario 2: Dynamic Bandwidth Walk (1-9 pkts/step)

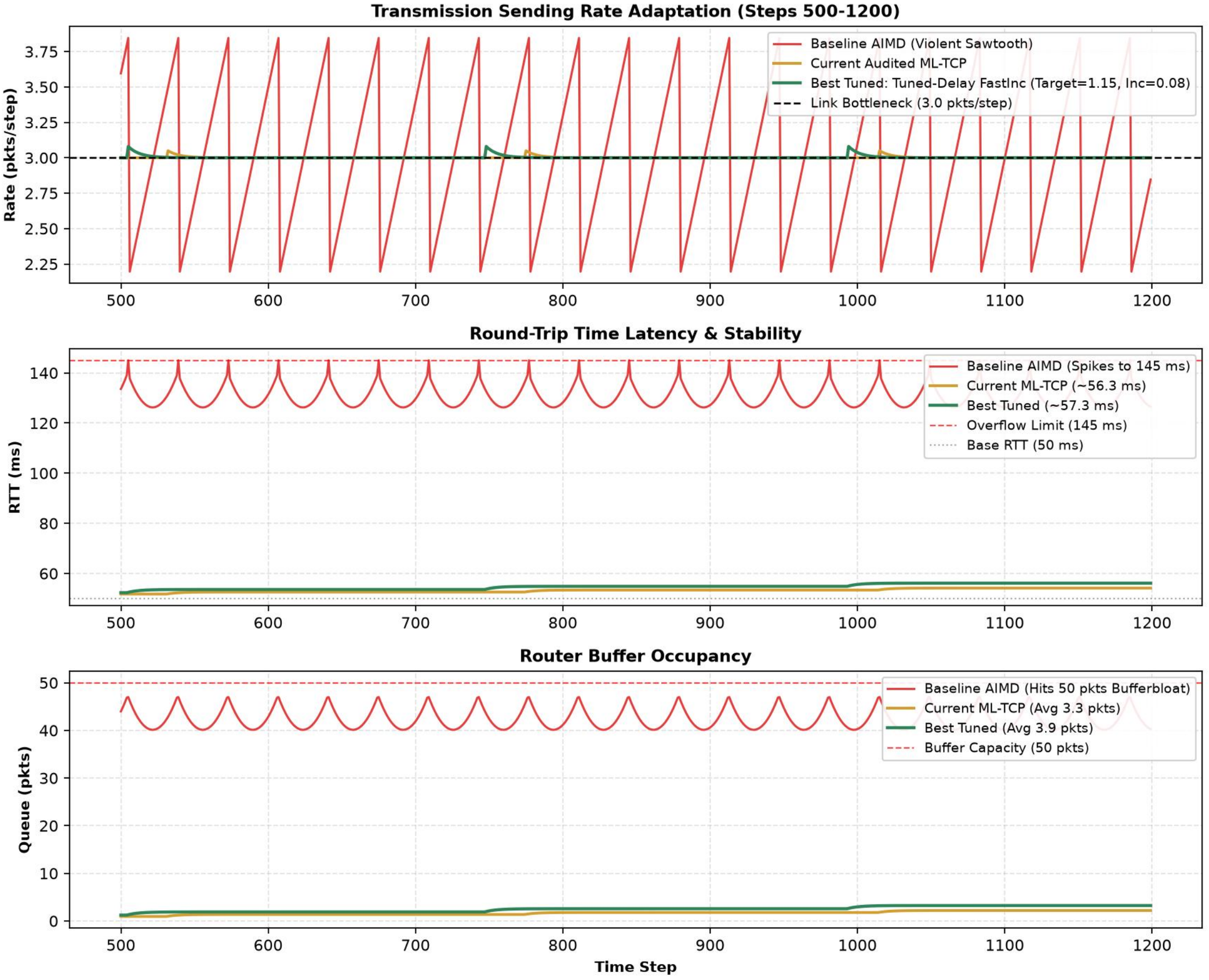


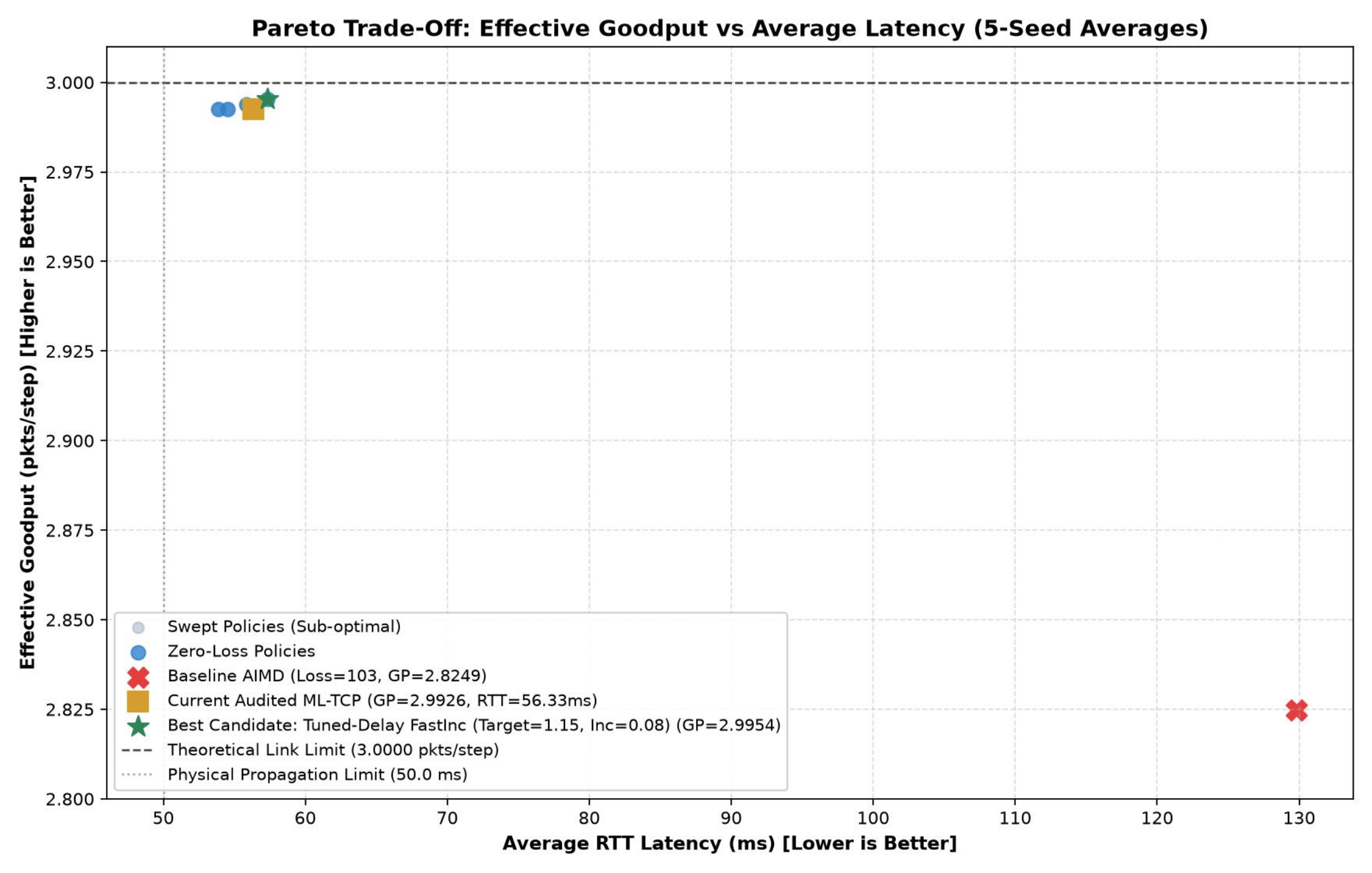
Scenario 3: Cellular Markov Fades with Deep Outages



Scenario 4: Capacity Step Change Collapse & Recovery




Transmission Sending Rate Adaptation (Steps 500-1200)
Rate (pkts/step)
Baseline AIMD (Violent Sawtooth)
Current Audited ML-TCP
Best Tuned: Tuned-Delay FastInc (Target=1.15, Inc=0.08)
Link Bottleneck (3.0 pkts/step)
Round-Trip Time Latency & Stability
RTT (ms)
Baseline AIMD (Spikes to 145 ms)
Current ML-TCP (~56.3 ms)
Best Tuned (~57.3 ms)
Overflow Limit (145 ms)
Base RTT (50 ms)
Router Buffer Occupancy
Queue (pkts)
Baseline AIMD (Hits 50 pkts Bufferbloat)
Current ML-TCP (Avg 3.3 pkts)
Best Tuned (Avg 3.9 pkts)
Buffer Capacity (50 pkts)
Time Step


Pareto Trade-Off: Effective Goodput vs Average Latency (5-Seed Averages)
Effective Goodput (pkts/step) [Higher is Better]
Average RTT Latency (ms) [Lower is Better]
3.000
2.975
2.950
2.925
2.900
2.875
2.850
2.825
2.800
50
60
70
80
90
100
110
120
130
Swept Policies (Sub-optimal)
Zero-Loss Policies
Baseline AIMD (Loss=103, GP=2.8249)
Current Audited ML-TCP (GP=2.9926, RTT=56.33ms)
Best Candidate: Tuned-Delay FastInc (Target=1.15, Inc=0.08) (GP=2.9954)
Theoretical Link Limit (3.0000 pkts/step)
Physical Propagation Limit (50.0 ms)